Observers Who Read Their Universe into Existence: Finite Consensus, an Executable Observer, and Standard Physics

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Abstract

Observer Patch Holography asks whether one self-recovering executable history can account for every public physical fact. Its central finite construction joins protected consensus records to a twelve-port carrier: the carrier has an exact three-directional slow response, and its complete response has the Standard Model gauge Lie type under stated endogeneity premises. A specified matter and memory implementation places native currents, a pixel-dependent lepton matrix and localized modes on common finite operators, with controlled memory limits and nonquadratic interactions. Compact opposite-degree textures have finite energy bounds, while their binding and interactions change the excitation spectrum. Quantum record identities use a supplied complex algebra-state representation. A centered refinement gives eight free massless Weyl valleys; a distinct spin construction has gapless continuum observables. Einstein dynamics follows from a realized common null balance, stress conservation and geometric entropy premises. The physical joins from these constructions to the observed particle spectrum, one interacting chiral field and responsive gravity are unproved. The closure proposal also frames questions of existence and experience without deriving phenomenal qualities. Frozen dispersion and instrument tests make specified parts falsifiable.

Keywords: foundations of physics; Observer Patch Holography; observer consensus; executable observer; self-reference; quantum records; gauge structure; Einstein field equation.

1 Introduction

1.1 Mission

The mission of OPH is a consistent and complete, formal and rigorous theory of reality. Its proposed complete history has no external starting point: it closes on itself. A successful theory must meet four conditions.

1. **Consistent.** The loop closes. Its fixed point exists, is constructed explicitly and is machine-checked, and a theory with a model contains no contradiction.

2. **Explicit.** The complete recovered specification fixes its consequences. Every design choice and calibration is declared; quantities claimed as independent predictions are not inserted through those inputs.

3. **Complete**, in the sense Einstein, Podolsky and Rosen gave the word [18]. Every element of physical reality has a counterpart in the theory, and every measured quantity is derived.

4. **Correct.** Derived values agree with measurement, and new predictions are frozen before they are tested.

The mission ranges from the existence of an inhabited history and the possibility of experience to the number of spatial directions, quantum statistics, forces, particles, gravity and cosmology. This range specifies what the proposed account must explain; it does not turn a finite model or a conditional theorem into a derivation of every item. In particular, a three-dimensional carrier response needs a physical position map, and an operational observer needs an additional account before any claim about phenomenal qualities follows. Table 1 gives these distinctions alongside the physical results.

OPH does not claim to be the only theory with these properties. The four conditions organize this paper. Consistency is tested through the closure hypothesis and the executable model (Section 2) and through the machine-checked results of Section 14. Specification and parameter ancestry are tested through the closure coordinates of Section 11.2. Completeness is the ledger of realized results and physical obligations in Section 12. Correctness is decided by the frozen predictions and tests of Section 13.

1.2 Observers first

Physics usually begins with a world and asks how observers fit into it. Observer Patch Holography (OPH) begins with observers and asks what world they can establish together. Its observers are finite. Each holds a patch of local state, reads a boundary it shares with its neighbors, keeps durable records, and repairs its state when records disagree; it accepts only repairs that preserve protected data. OPH identifies an objective fact with a record that survives every authorized comparison. Quantum probability, Lorentzian kinematics, gravitational dynamics and gauge structure enter standard physics through distinct mathematical inputs [19, 20, 21, 55, 62]; OPH asks whether they are different public readouts of one finite observer architecture. Space, time, matter and classical objects are effective descriptions of what observers can measure and jointly interpret; no observer-independent continuum is assumed beneath them. In this interpretation, spacetime geometries and quantum field theories, including any conformal limit, describe relations established by compatible observer records; they are not additional entities posited to exist before or independently of those observer relations.

The intended argument runs from specified repair operations to compatible joint records, and from those records to a common operational description of space, time and matter. A position is what the spatial readouts locate; a duration is what calibrated clocks compare; a particle is a persistent excitation with the observed transport, charges and interactions. The target is one such description with three spatial directions and local time, including its quantum fields and gravitational response. The finite constructions below establish parts of this chain; the common physical $3+1$ limit is the remaining conclusion to be demonstrated, not an input hidden in the word "observer."

OPH makes one proposal about the whole, closure: one complete history is both a timeless self-referential structure and its own internal computation, and observers inside it recover its specification and realize it (Section 2). The executable-model hypothesis is the concrete and testable

form of that proposal: one specified local repair and memory process must reproduce the observed quantum fields and their common gravitational description. It names one carrier architecture, one local repair and memory law and one matter mechanism, and it requires one source process to supply records, clocks, fields, matter, energy and responsive geometry together. This paper presents OPH as that single theory, organized by its dependencies: each layer states a hypothesis, proves finite theorems on displayed premises, and passes a conclusion upward only when the executable model realizes the premise on the same process.

1.3 Scope and order of the argument

The strongest joint finite result uses one carrier architecture. Its twelve-port response has an exact rank-three slow band, while complete connected response with internally implemented rechartings has $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$ Lie type. The specified matter lift and memory rule place those currents, pixel-dependent coefficients and localized modes on common finite operators. This connection does not identify the band with physical space or the modes with measured particles. Section 8 gives the operator construction.

Section 9 distinguishes the exact interacting finite algebra from controlled spatial limits. Its energy, Gibbs response and complete memory records refer to the same Hamiltonian. The separate fermionic and spin refinements have stated free or interacting domains; none is identified with a complete observed chiral spectrum. The Einstein theorem consumes common geometry and entropy premises beyond these finite identities. Section 12 collects the physical requirements that each implication needs. The specialist papers contain the detailed operator constructions and proofs [12, 11, 9, 63].

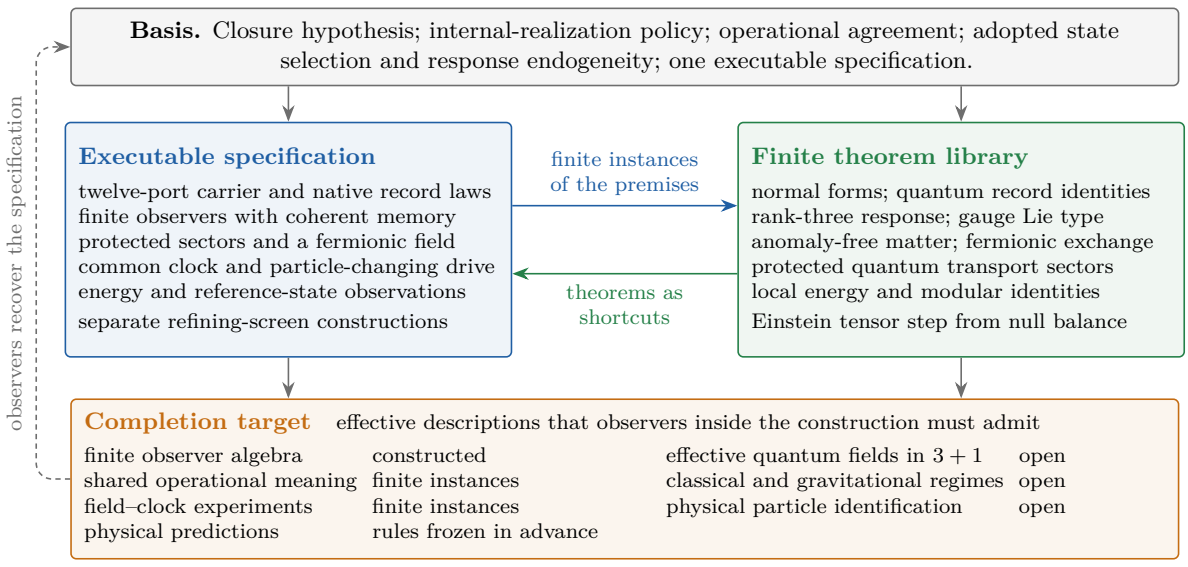


Figure 1: How the parts of OPH fit together. The basis fixes one executable specification and the premises of the finite theorem library. The running specification supplies finite instances of those premises, and a theorem replaces simulation wherever its premises are decided on the specification's finite objects. Both feed the tower of the executable-model hypothesis, whose layer status Section 12 gives. The dashed arrow is the closure loop of Section 2: observers recover the specification from their records.

Finite theorems, complete source executions and physical interpretations answer different ques-

tions. We state each result's domain and supplied inputs where it is used; the detailed claim classes and verification methods are collected in Section 14.

"Source-executed" means that an operation is compiled and run using the declared native preparation, repair, memory and readout services, with its complete outputs and resources accounted. "Source-only" concerns the provenance of its parameters: no empirical physical input occurs anywhere in their dependency chain. These are different properties. A source-executed field may use empirically calibrated Yukawas, mixing parameters or clock units; reproducing those inputs verifies the implementation, not an independent prediction.

Definition 1.1 (Physical realization map)**.** A physical realization map assigns finite public records, event data, currents, modular generators, and scales to physical records, spacetime events, laboratory currents, clocks, and units, preserving the algebraic operations, overlap restrictions, causal order, normalizations, and refinement relations consumed by the conclusion in question. A resemblance of spectra, dimensions, or symmetry names is insufficient.

Order of the construction. Section 2 states the hypothesis and its additional premises. Section 3 defines the carrier and both native record laws before any calculation uses them. Public quantum records, spatial response and executable observers then provide the interfaces needed for fields and matter (Sections 4–8). Section 9 assembles the protected-sector, field, clock and energy constructions on explicitly joined carriers. The Einstein and cosmological arguments follow with their own hypotheses (Sections 10–11). Finally, Section 12 consolidates the realized results and physical obligations; Sections 13–14 give tests and reproducibility. The companion construction and component papers carry the full implementation contracts and longer proofs [63, 1, 4, 5, 6, 8, 11, 9, 7, 13, 14, 17, 2].

2 Closure and the executable-model hypothesis

This section states the interpretation and acceptance criterion: closure, the internal-realization policy, the additional state-selection and response-endogeneity principles, and the agreement condition, then explains why closure is tested through an executable model.

2.1 The closure hypothesis

The universe is modeled as a timeless, closed, self-referential structure in which the description and the described system are one object. Let $\mathfrak{U}$ be a complete inhabited history, let $D(\mathfrak{U})$ be an account of its architecture recoverable through completed records within it, and let R be an admissible realization of that account constructed inside $\mathfrak{U}$. The closure condition is

$$R(D(\mathfrak{U})) \;\cong\; \mathfrak{U}, \tag{1}$$

where $\cong$ is agreement in the observable and interventional structure: the records used to recover the description, and the responses of the realization to every admitted intervention, fit the world from which the account was recovered. Written as a fixed point of a universe-level closure operator, the condition reads $T(\mathfrak{U}) = \mathfrak{U}$. Observers infer the specification of their world, build hardware within it, and run that hardware; its execution realizes the same history, including its builders and their hardware. The identification of the construction-side and inhabited-side readings is the hypothesis. In this sense the structure reads itself into existence.

Within this hypothesis, no external computer runs the history and no additional time exists in which the whole is played. Host runtime used to check a model is part of our method of investigation,

not an extra time coordinate of the universe described by that model. Repair order is a partial order on commits, an observer history is a chain of completed records, and that order is invariant under strictly increasing regradings, so affine time and proper time require their own calibration. Fixedness of the complete history permits changing states within it: the equation concerns the whole history and identifies no last instantaneous state with the universe.

Self-reference earns content only when the return condition restricts the possibilities. On the real line $x \mapsto x+1$ has no fixed point, $x \mapsto x$ fixes every point and $x \mapsto (x+2)/2$ fixes exactly one. A candidate therefore has to specify the history domain and the action of T; existence, consistency and the identification of our universe with a fixed point are questions the equation leaves open. Self-reference alone guarantees neither consistency nor mathematical completeness. Uniqueness is an additional question when it is claimed; it is not a prerequisite for a specified construction to work and match observations. Two finite results give the condition exact content on declared branches. The packet-quotient closure theorem proves that on the declared finite consensus branch the closure map is an affine idempotent whose fixed points are exactly the packets supported on consensus normal forms [6]. The universe-level record-closure equation $N = \log M_0(\mathfrak{U}_N)$, with $\mathfrak{U}_N$ the regulated observer system at capacity budget N and M_0 the correctable-code capacity of its terminal public record, is posed exactly, and the all-rung arithmetic proves that the bounded completion class defined by base agreement, positivity and the carrier bound does not identify it [8].

2.2 No outside

A self-contained realization must account internally for its operations and information access. This motivates a justification policy and two further constitutive principles; it does not logically derive those principles.

Justification policy. A premise is justified operationally when observers recover it and the specification implementing it runs and reproduces their measurements. The carrier, observer algebra, repair law and closure coordinates must meet that test. Their specification can be reconstructed from records or proposed as a design. In particular, using a twelve-port carrier does not require a theorem that every possible realization must have twelve ports. A construction choice needs no uniqueness theorem unless uniqueness is claimed. Closure neither exempts a candidate from empirical tests nor proves its antecedents jointly realizable.

State-selection principle. On the declared feasible family, reference, observer cover and positive weights, adopt the information projection minimizing $\sum_p w_p D(\rho_p \| \tau_p)$. This is premise A3, motivated by least-informative inference but separately adopted when used as a physical selection law. The closure equation supplies no entropy functional or reference; even its identity-map specialization fixes both uniform and nonuniform states. Under A3, every minimizing local density has the union of feasible supports, and a feasible failure witness $b > 0$ gives an explicit selected-probability floor [9].

Response-endogeneity principle. The complete-response branch additionally requires each proper carrier recharting to be implemented by the connected response group, up to a unitary centralizer acting trivially on the response algebra. This is the endogeneity clause of A2, not a consequence of internal implementation alone. For example, twelve diagonal phase generators on $\mathbb{C}^{12}$ admit internal port permutations in $\mathrm{U}(1)^{12} \rtimes A_5$. These permutations act nontrivially on the abelian current algebra, whereas its inner automorphisms are trivial. This closed operation system

fails the stronger endogeneity premise; it shows why that premise must be supplied. Theorem 8.1 uses it to select $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$ under the stated complete-response hypotheses.

2.3 Agreement and the meaning of "our"

The justification policy fixes the operational test. Agreement says what counts as shared. Observers assign the same operational meaning to accepted data on shared boundaries, naturally under every restriction, recharting and refinement, and a record is public when it survives every authorized comparison (Theorem 4.4). "Our universe" is the world that this public record describes. Agreement in this sense is a definition and an acceptance criterion: it requires interpretation maps fixed independently of validation outcomes and preserving the declared joint and conditional statistics, including quantum correlations. It does not require a noncontextual table of outcomes for incompatible unperformed measurements. Agreement here is compatibility under specified comparisons and interventions, not a vote among observers or assent to a convenient interpretation. Its content includes the joint probabilities and the response to changed preparations. Calling space, time and matter operational descriptions does not make these constraints optional: once established by compatible records, they are public physical facts. The equations describe those relations and do not by themselves assert an additional material substrate beyond the specified process. The theorem library calls the interpretation clause of this condition A2. The basis reference distinguishes the closure hypothesis, agreement, adopted selection/endogeneity principles and executable specification, with the library's premise statements and their dictionary [94].

2.4 An entry point without a starting point

The exposition of this paper enters the loop at the executable specification. That choice reflects where observers stand, holding records and inferring, and it makes the account concrete. Closure states the internal realization contract; the adopted principles and specification then carry their own mathematical and empirical tests. The order of exposition is a logical entry point, not an event before the universe or a privileged temporal beginning of the self-referential loop. A pair of mutually determining equations such as $x = (y + 2)/2$, $y = (x + 2)/2$ fixes $x = y = 2$ with no first value from which the other is manufactured; a calculator may approach the pair by iteration and someone may find it on a particular day, and neither belongs to the reason the answer is two.

The closure equation relates parts of one timeless object; it posits no backward signal in an external time coordinate. Observers recover the specification and later construct it in their own record order, and both stages belong to the same fixed point. In the terminating consensus theorems, repair descent supplies a well-founded order on the admitted updates for proving termination and comparing schedules; it is neither an observer's experienced time nor a universe-level clock.

What is reconstructed and what is constructed are two roles within this proposal. The observer who infers a law and the hardware that realizes it both belong to the candidate history. Finding that specification in a laboratory or checking it by computation occurs in the investigator's record order; neither supplies a temporal origin outside the history. If several specified closures satisfy the same tests, the results warrant that observational class without a uniqueness claim.

2.5 From closure to an executable model

The executable observer gives the condition its finite meaning. A realization differs from a description because it carries the described dependencies under intervention: for a described update τ_a, realized operation F_a and fixed readout p, the one-step condition $p(F_a(s)) = \tau_a(p(s))$ extends by induction to every finite admitted sequence. A playback can match every frame of one history and

fail this test as soon as a retained state is changed. The controls of Section 6 perform exactly this test on the running observer: disconnecting the detector or shuffling its bit with the same frequencies removes the feedback response, while the record-controlled execution keeps it.

Closure alone does not imply that a particular carrier, record law or initial state yields Standard Model physics. The executable-model hypothesis is the stronger claim that one simple, fully specified law does, and it is the form in which closure can fail. Its carrier architecture, refinement and composition, record instruments, memory law, scheduler, preparations and available controls are stated before testing. A failed specified law rejects that law and its tested interpretation; it reaches the wider proposal only when the rejected choice is proved necessary and exclusive. Changing the law, memory, state or decoder defines a new candidate.

2.6 Specification and theorem premises

The architecture, record instruments, memory law, scheduler, preparation services and allowed interventions form a declared specification. Observers may infer or design these ingredients; mathematics determines their consequences, and comparison determines whether the resulting process matches observed physics. Declaring a primitive fixes the model under study. Its consequences require proofs, and its physical adequacy requires tests; a specified input must remain distinguishable from a consequence derived from other inputs. The carrier, memory and control laws need not be uniquely selected by closure to be legitimate inputs. Their implementation and resource use must, however, belong to the process being studied.

Three premise families recur. A1 specifies the carrier, federation and response structure. A2 adds agreement on shared records and inner implementation of the complete response algebra. A3 adopts a least-informative state-selection principle on the stated feasible family. Closure alone proves neither the entropy minimizer nor the inner implementation assumption. Later theorems state which of these inputs they use and which additional hypotheses their conclusions require.

Quantum executions below use finite-dimensional complex Hilbert spaces, positive normalized density operators and completely positive instruments whose complete effects sum to the identity. These are explicit quantum premises, not consequences of classical normal-form uniqueness. A faithful reference is a strictly positive density operator $\rho > 0$; its modular flow is $\sigma_s(A) = \rho^{is} A \rho^{-is}$. The dimensionless parameter s requires a separate implemented calibration before it is read as duration. For fermionic fields the canonical anticommutation relations (CAR) are $\{c_i, c_j^\dagger\} = \delta_{ij} I$ and $\{c_i, c_j\} = 0$; the declared Fock representation fixes the exchange signs used later.

The physical interpretation also fixes a domain: one prepared history, a family of interventions, or every allowed input state. These are inequivalent requirements. Complete quantum instruments must retain the reference systems and all outcomes needed for the chosen domain; a record history need not assign simultaneous values to unperformed incompatible measurements. Likewise, a stationary probability law, an instantaneous repair fixed point and the closure of an entire history are distinct. Host runtime, repair counts, modular parameters and measured duration are related only by an implemented clock calibration. The operational observer below is defined by access and record interfaces, without an additional claim about consciousness.

3 The observer and the native repair law

The model rests on two objects: an operational observer and a twelve-port carrier with its native repair law. We define their interfaces and update rules before constructing the observer that runs on them.

3.1 Patches, records and repair

Definition 3.1 (Observer patch)**.** An observer patch is a finite object with local state, an observable boundary map, durable records, readback, and a set of authorized repair moves. A patch sees a fragment of the world: its own state and the boundary data of its authorized overlaps.

Definition 3.2 (Overlap and repair)**.** For patches x_i, x_j with an authorized overlap $e = (i, j)$, both induced boundary records must agree for the pair to hold public data. A repair move changes local state while preserving protected readout. A configuration is a normal form when no authorized repair applies.

An observer in this sense is defined by its access and record interfaces. It reads, writes and retains records, and it uses retained records to choose its next operation. Nothing in the definition refers to a background space, a global clock or a pre-assigned list of particles. Those enter later as readouts that observers can share.

3.2 The twelve-port carrier

Each local carrier has twelve primitive boundary ports with oriented icosahedral incidence: thirty seams and twenty faces, five seams at each port, every seam on two oppositely oriented faces. This is finite combinatorial incidence, not an embedding of fundamental objects in an pre-existing physical space. Coordinates used to draw the carrier or compute its group representations describe that finite object; physical position and metric enter through operational reconstruction. Carriers federate through typed seam algebras with coherent triple overlaps into a nerve carrying a degree-one bridge to an oriented spherical support. The theorem library calls this carrier, federation and response layer A1. Within the closure hypothesis of Section 2 it is the part of the specification that observers recover from their records; every result below displays its dependence on it. The local carrier Gram is recovered from native current records, while cross-carrier routing in the executed federations uses a supplied cell-centre and tangent-frame convention.

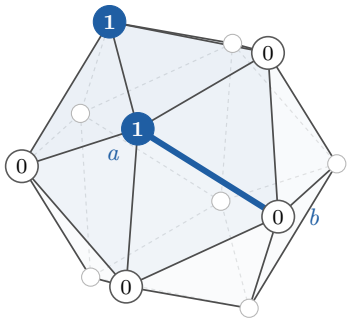
Two exact features of the carrier enter later constructions. First, the proper rotation group of the icosahedral packet is the alternating group A_5, with sixty rotations acting on the twelve ports, and the real port-coefficient space decomposes without multiplicity as

$$P_{12} \cong_{A_5} \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}. \tag{2}$$

On the six-dimensional golden sector the spectral projectors have entries in $\frac{1}{20}\mathbb{Z}[\varphi]$, $\varphi = (1 + \sqrt{5})/2$, each is the entrywise Galois conjugate of the other, and both triplets are irreducible real representations of real type; the projectors, characters and irreducibility certificates are machine-checked [50, 1]. Second, record keeping is integer valued: a write appends $+1$, a retraction appends -1, and readback sums atomic events per port. A conservative repair transfers one unit across a seam whenever the oriented mismatch has magnitude at least two. With $V(x) = \sum_i x_i^2$, each such repair strictly decreases V by $2(d-1)$ for mismatch $d \geq 2$, so repair terminates. The native repair law of Section 3.3 also moves load to the floor and ceiling of the endpoint mean, but at difference one it waits or swaps by its coin, so the native process keeps moving while its quadratic cost stays fixed.

3.3 The native repair law

The executable observer runs on one carrier with twelve nonnegative integer loads of total two. There are 78 such configurations, and the 66 binary configurations form an invariant shell. For an



twelve ports, thirty seams, twenty faces; a binary load of total two (filled ports)

$d = x_a - x_b, \qquad j = \frac{1}{2}\big[d - (2c-1)(|d| \bmod 2)\big]$

d	$c=0$	$c=1$	effect of the attempt
2	1	1	balance
1	1	0	swap or wait
0	0	0	no move
-1	0	-1	wait or swap
-2	-1	-1	balance

The move j carries j units from a to b, and the fair coin c reads no load. At $|d| = 2$ the attempt lowers $V = \sum_i x_i^2$; at $|d| = 1$ it keeps V fixed, so the native process keeps moving. The seam drawn has $d = 1$.

Figure 2: The twelve-port carrier and its native repair. Left: the icosahedral incidence of the twelve ports, with thirty seams and twenty faces, in an orthographic view with back edges dashed; the filled ports carry a binary load of total two, and the drawn seam (a, b) has $d = 1$. Right: the seam rule for every mismatch the total-two carrier admits.

oriented seam (a, b) and a fair placement coin $c \in \{0, 1\}$ (Figure 2), the native repair law moves

$$d = x_a - x_b, \qquad j = \tfrac{1}{2}\big[d - (2c-1)(|d| \bmod 2)\big], \qquad y = x + j\,(e_b - e_a), \tag{3}$$

so load is conserved and the signed current j is the native observation. Conservation, one-step balance, a tie-break that reads no load and endpoint symmetry characterize this rule; on the total-two carrier they fix only the tie-break coin. Its quantum version is a record law, and seam locality, the classical law and the declared local number-phase and endpoint-swap covariance, a local phase symmetry of the repair channel, leave a convex family of them, each a table of five coherence multipliers [65]. The two laws used below have the same load update (3) and differ in which information they record. Write $p_e(x) = (x_a + x_b) \bmod 2$ for the parity of the seam's total load. Law C records the pair (c, j); law E records (c, j, p_e). Their Kraus operators are

$$L^C_{c,j} = \frac{1}{\sqrt{2}} \sum_{x:\, j(x,c)=j} |y(x,c)\rangle\langle x|, \tag{4}$$

$$L^E_{c,j,p} = \frac{1}{\sqrt{2}} \sum_{x:\, j(x,c)=j,\ p_e(x)=p} |y(x,c)\rangle\langle x|. \tag{5}$$

Each complete instrument preserves total probability. E exposes an extra parity record and therefore removes coherence between the corresponding parity classes when the record is ignored. It changes no classical load or current probability. C is the convention used for the primary construction; E provides a specified alternative and control. A current-only coherent implementation gives the most coherent member C_0; adding a coin readout gives C, and adding seam parity gives E. Classically forgetting a written coin is not the coherent C_0 implementation. E and the seam-total recorder are the two idempotent members of the classified family. Under a common definite-current record interface all members share every classical current statistic. One coherent interference experiment reads the multiplier g that separates C, E and the most coherent member (Section 13), and nine source-executed measurements read every real coordinate of the table (Section 6). The quantum relaxation of the channels obtained by discarding the source record, the forgotten-source channels, is known exactly. C and E have quantum spectral gap $(5 - \sqrt{5})/60$ on the binary shell and on the full carrier, their slowest binary coherences decay by the sharp factors $5/6$ per attempt

for C and 7/10 for E, the binary-shell channel becomes entanglement breaking after finitely many attempts (134 attempts suffice by an explicit certificate), and on the full carrier it never does: an explicit partial transpose stays negative at every attempt count [64, 66]. The endpoint instruments also fix a local action. On binary operator space $H_{\text{native}} = \sum_e (I - \Phi_e)$, summed over seams with Φ_e the unread C or E channel, is positive and local, with an exclusion exchange term and coherence-killing coefficients set by the record law; the uniform discrete transfer is $I - H_{\text{native}}/M$ for M seams. It is a Liouville operator on operators, not a Hamiltonian on carrier states, and the executed channel strictly attenuates a coherent mode, which rules out unitary conjugation on this carrier. At fixed selected-pair sum each classical repair minimizes the pair's quadratic load cost, so $\|x\|^2/2$ falls by $(d^2 - (|d| \bmod 2))/4$ per repair [78, 64].

The gap and the arrow do not depend on the record law. Every admissible coherence table has the uniform binary state as its unique stationary state and the gap $(5 - \sqrt{5})/60$; the linear load statistics along the rank-three port vectors of Theorem 5.1 attain it, while every coherence between distinct load configurations contracts by at most $43/60 + 2\sqrt{2}/15 < 1 - (5 - \sqrt{5})/60$ per attempt, so the longest-lived datum of a carrier is a classical position-like readout and equilibrium selects no record-law coordinate [65]. On the binary shell each native repair is the fibre-resampling map of the theorem library for the uniform reference: it resamples the seam orientation uniformly within the fibre of its repaired visible datum, which minimizes relative entropy over that fibre, while a collision is resolved by deterministic descent. The library's second law and modular Clausius inequality therefore hold for every native attempt on the binary shell [8]. One such projection is idempotent; the declared uniform schedule over noncommuting seams supplies a sustained arrow and contracts every input to the uniform state at the gap rate, and passive repair never realizes the library's third map, the modular flow of a faithful reference.

4 Public records and completed-record identities

The repair mathematics establishes when local agreement reaches global consistency and when nonzero cycle holonomy leaves a residue that local repair cannot remove. Two observers who share a boundary can disagree about what it shows. If repair stopped in different places depending on who acted first, the boundary would carry no fact of the matter. If it always stops in the same place, the final record is a public fact that no single observer owns. This section states when the second case holds, and then shows that completed records satisfy the familiar quantum identities on the declared complex algebra and state domain below.

A completed finite configuration x is a fixed point of the named repair-and-agreement map,

$$\mathcal{R}_{\text{cons}}(x) = x.$$

This map is distinct from the universe-level closure operator of Section 2. Fixed-point existence, observable determination and schedule independence are separate questions, and the following results isolate them [4, 6].

4.1 Observable normal forms

Definition 4.1 (Observable quotient system)**.** An observable quotient system is a tuple $\mathfrak{S} = (Q, C, \mathcal{B}, B)$, where Q is a set of configurations, $C \subseteq Q$ is the consistent subset, $\mathcal{B}$ is a set of protected records, and $B : Q \to \mathcal{B}$ is the record map. For $b \in \mathcal{B}$, write $C_b = C \cap B^{-1}(b)$.

The fiber C_b is unrealizable, reconstructing, or ambiguous according as its cardinality is zero, one, or greater than one. This trichotomy avoids assigning a state to a record that has either no consistent extension or several observationally indistinguishable extensions.

Theorem 4.2 (Canonical observable normalizer)**.** *The following statements are equivalent:*

(i) $B|_C$ *is injective;*

(ii) there is a unique map $\mathcal{N} : Q \to C \sqcup \{\bot\}$ *that preserves* B *on its* C*-valued outputs, fixes every point of* C*, is constant on each* B*-fiber, and returns* $\bot$ *exactly on fibers with no consistent extension;*

If $B(Q) \subseteq B(C)$*, the map is a total idempotent retraction* $Q \to C$*.*

Proof. Injectivity makes every nonempty C_b a singleton, which defines $\mathcal{N}$ and forces all its properties fiber by fiber. Conversely, if $c, c' \in C$ have $B(c) = B(c')$, the fixed-point and fiber clauses give $c = \mathcal{N}(c) = \mathcal{N}(c') = c'$. □

The normalizer is defined by observable fibers, independently of any repair schedule; a repair relation implements it when the relation preserves B, reaches the consistent set and satisfies a liveness condition. Approximate versions control the normalizer where exactness fails, through an optimal two-output distance bound and summable naturality defects along refinement towers. Canonicity carries no promise of cheapness: for succinct Boolean boundary systems, deciding whether a protected record has a consistent extension is NP-complete, and deciding injectivity of the record map is coNP-complete [4].

For the canonical single-site repair, every adaptive run is eventually constant because each nonstuttering attempt strictly lowers the mismatch count; pathwise weak fairness makes the stable state a normal form, and completeness with confluence sends every weak-fair run to the same public fixed point. These results select no scheduler and provide no stochastic hitting law, rate, physical clock, or refinement theorem. Attempts obey a sharp capacity law. A scheduler can delay the first genuine repair for any prescribed number of attempts. When every reducible record sees a genuine change among the next $q + 1$ attempts, a bounded-waste upper horizon of $(q + 1)$ times the initial mismatch count holds, and an independent-defect family attains the work-conserving case exactly at $q = 0$. The input-independent compiler produces a fixed federation with immutable input ports; genuine node repairs decrease a dependency rank, and pathwise weak fairness gives stable consensus and the correct output. Emission-order round robin finishes within its first n attempts, and the single-consumer structure bounds every accepted path by $n(n + 1)/2$, with an emitted family attaining $\Theta(n^2)$. General well-formed circuits instead admit $2^n - 1$ accepted steps, attained by a fanout chain, where n counts the emitted nodes. These statements concern canonical node repairs: the weak patch-frame relation can stutter fairly forever, and no physical scheduler, clock, energy or hardware cost follows from these operation counts [4, 6].

4.2 Public records from local repair

For a prepared batch of support-closed transactions, join τ and σ in the conflict graph when $W_\tau \cap (R_\sigma \cup W_\sigma) \neq \varnothing$ or $W_\sigma \cap (R_\tau \cup W_\tau) \neq \varnothing$, and iterate to a fixed point: compute semantic support inside every aggregate, rebuild the conflict graph on the expanded read and write sets, merge every connected component, and repeat until the partition and all aggregate supports are unchanged.

Proposition 4.3 (Transactional local diamond)**.** *Let an accepted repair be an aggregate transaction with read set* R_τ*, write set* W_τ*, a read snapshot, and a snapshot-determined payload. Assume:*

(i) every read set contains the semantic dependency closure of its write set, and every affected acceptance functional is revalidated at commit;

(ii) the terminal closure above exists, and each of its connected conflict components has one coherent canonical aggregate payload;

(iii) a prepared component whose snapshot remains valid survives commits of independent components; and

(iv) accepted commits preserve the protected record and strictly decrease the integer descent functional.

Then every one-step quotient peak $t \longleftarrow s \longrightarrow u$ has a quotient state v with $t \longrightarrow v \longleftarrow u$.

Proof. Two different first steps cannot originate in one conflict component, since that component has a unique aggregate payload, so they originate in distinct components of the terminal closure. Rebuilding the graph after every aggregate support expansion makes their final read and write sets disjoint, so neither commit changes the other's read snapshot, and semantic dependency closure ensures neither changes an acceptance functional consumed by the other. Revalidation and the surviving component rule make both second commits legal, and with snapshot-determined payloads and disjoint writes the two application orders agree; protected-record preservation carries the equality to the physical quotient. □

Theorem 4.4 (Consensus normal form)**.** *Suppose the accepted repair relation satisfies the hypotheses of Proposition 4.3, so that it has the one-step diamond property, strictly decreases the quadratic descent functional V of Section 3, and is repair complete, meaning that its normal forms are exactly the consistent states. Every maximal asynchronous repair schedule from one initial state then terminates at the same consistent quotient normal form. The induced global repair map is idempotent, fixes consistent states, and preserves the protected record.*

Proof. Quadratic descent makes the accepted relation terminating. Proposition 4.3 makes it locally confluent, so Newman's lemma [43] makes it confluent. A terminating confluent relation has a unique normal form below each source, and repair completeness puts that normal form in the consistent set. Applying the normalizer a second time does nothing, and protected-record preservation holds along every accepted edge. □

Atomic commits, disjoint write sets, or termination alone do not imply the local diamond; countermodels omitting semantic dependency closure, canonical aggregation or the surviving component rule are retained with the finite receipt. This is the precise sense in which a public fact is what survives every authorized comparison: the public object is the protected quotient record, independent of the order in which observers repaired.

Repair also has a residue that it cannot remove. On a fixed graph with positive mismatch weights and fixed edge offsets, nonzero cycle holonomy forces a strictly positive minimum mismatch, and no repair of the vertex assignments eliminates it while the offsets are fixed [6]. The matter hypothesis uses this frustration as a candidate mechanism (Section 8.3).

The law runs at scale. The public run uses the level-six cell rung of the geodesic icosahedral refinement tower: 81,920 exact twelve-port carriers with one declared gluing of ports to neighboring cells, 2,580,480 seams in all, and the seam-mean retraction as repair primitive. Sixteen shuffled schedules from one seeded load terminate at the balanced class with no accepted move raising V, and all sixteen return one quotient hash equal to the component multiset fixed by the initial totals. A standalone verifier that imports no simulator code rebuilds the component multisets from the primitive arrays, replays one schedule from its seed and recomputes every hash [2]. The level-ten rung, with 20,971,520 carriers and 660,602,880 seams, terminates all sixteen schedules at

the balanced class in 84 to 100 sweeps with one quotient hash fixed by the loads. Across the rungs from five to nine, one initial load per rung gives an excess-energy contraction between 0.81 and 0.83 per sweep, and the sweep counts to the balanced class grow slowly with the carrier count, from 51 to 79 at level six to 84 to 100 at level ten; these finite readings prove no all-size mixing or termination-time bound. On the sixteen level-six schedules the number of unit transfers to the balanced class lies between 924,832 and 925,708, about 1.25 times half the initial ℓ_1 deviation of the loads from their mean. The first-settling configuration retains spatial information that its public component multiset does not encode: at level ten the initial-to-terminal amplitude responses of equal-count ancestral groups of sizes 1,024, 4,096 and 65,536 are 0.90758, 0.95232 and 0.98983, with conditional schedule-noise fractions 1.241, 0.594 and 0.143 percent. Stationary continuation adds a constructive fluctuation result: for fair integer repair on a connected weighted graph with independent edge clocks of total rate one the fixed-occupancy law is reversible, centered loads have lagged covariance $\kappa e^{-|t|L_G/2}\Pi$ in the edge-clock time t, with L_G the weighted graph Laplacian, $\kappa = r(p-r)/(p(p-1))$ for r raised occupancies on p ports and Π the projection off constants, and fixed linear readouts obey a martingale central limit theorem [7].

The same records let observers recover the specification itself. At carrier scale, recovery from event logs under the declared probe and operation grammar returns the twelve ports, thirty seams, inverse pairing, automorphism and rotation groups of orders 120 and 60, mean rule, schedule, rank-three Gram and Lie split $1+3+8$. Re-instantiation preserves the invariant vector for two iterations, and tetrahedral and octahedral controls recover their distinct specifications [2]. Complete federation gluing is not identified by the retained routing logs: within the declared port-gluing interface a two-edge switch preserves every carrier degree, occupied port and source traversal while changing the carrier graph's triangle count from 14,000 to 13,990, and native executions reproduce identical bytes for both. The alternative is not asserted to belong to the canonical geometric family. Full-gluing recovery requires further observations or a justified restriction of the candidate class.

4.3 Completed records and quantum identities

Once a compare, write and verify slice has completed, its accessible events are represented in a finite-dimensional complex C^*-algebra with a positive normalized state and completely positive instruments. This algebra-state representation is an explicit input; the consensus theorem identifies which record is public and does not by itself select complex quantum theory over every alternative probability model.

Theorem 4.5 (Finite public-record identities)**.** *Let P_E be the projector of a completed public event E and ρ the normalized state of the completed record surface. The event weight and nonzero-weight update*

$$\Pr(E) = \operatorname{Tr}(\rho P_E), \qquad \rho \,|\, E = \frac{P_E\, \rho\, P_E}{\operatorname{Tr}(\rho P_E)} \tag{6}$$

define a normalized probability measure on every orthogonal event partition and a normalized positive state on the post-event corner. For self-adjoint dichotomic observables A_0, A_1 and B_0, B_1 in commuting record subalgebras, with $A_i^2 = B_j^2 = I$, the combination $S_{\mathrm{CHSH}} = \operatorname{Tr}[\rho\{A_0(B_0 + B_1) + A_1(B_0 - B_1)\}]$ obeys $|S_{\mathrm{CHSH}}| \le 2\sqrt{2}$ [24, 25]. The bound is attained exactly on the committed slot-split interface by the maximally entangled state with Pauli and quarter-rotated projections, while every state diagonal in the record basis with diagonal observables valued in $[-1, 1]$ obeys $|S_{\mathrm{CHSH}}| \le 2$.

Proof. Positivity of ρ and $0 \le P_E \le I$ give $0 \le \operatorname{Tr}(\rho P_E) \le 1$, and $\sum_k P_k = I$ normalizes every partition; for nonzero weight $P_E \rho P_E / \operatorname{Tr}(\rho P_E)$ is positive and normalized. For $\mathcal{B} = A_0(B_0 +$

$B_1) + A_1(B_0 - B_1)$ the commuting-subalgebra calculation gives $\mathcal{B}^2 = 4I - [A_0, A_1][B_0, B_1]$ [26], so $\|\mathcal{B}\| \le 2\sqrt{2}$. The saturating witness and the diagonal bound are exact finite calculations [5]. □

At the level of weights the Born form is derived. Every valuation on the effects of the finite surface that is nonnegative, normalized and additive on coexisting effects equals $E \mapsto \mathrm{Tr}(\rho E)$ for a unique density operator ρ, in every finite dimension including two, where the projector-only Gleason theorem fails, and the derivation consumes no continuity axiom [22, 23]. The projectors alone do not select the update: an explicit swap-twisted instrument has the same effects and different updates. Kraus form with induced effect P together with invariance on the states certain of P is equivalent to being the Lüders map [29, 5]. The same surface carries the copying obstruction: if one ancilla-free isometry copies two sharp states from a common blank, preservation of inner products forces their overlap z to satisfy $z^2 = z$, so two distinct alternatives copied by one device are orthogonal [27, 28]. Finite webs do not by themselves supply the valuation hypothesis: a source-produced complex-tomographically-complete effect and instrument web with full coexistent-effect additivity is the missing step [5].

5 Three spatial directions

The carrier incidence fixes the dimension of the carrier's own position readback: the slow band is the three-dimensional irreducible representation of the icosahedral group, so the number three is a property of the committed carrier.

Theorem 5.1 (Intrinsic three-dimensional carrier completion)**.** *Let A be the twelve-port adjacency, let $M = I - (5I - A)/60$ be the uniform scalar repair mean, and take the complete equally weighted centered probe census. The normalized infinite-response kernel converges to the intrinsic Gram $G = 4P_{\mathrm{slow}}$, of rank three. The twelve integer port loads have antipodal-odd quotient $\mathbb{Z}^{12}/\{x : x_p = x_{-p}\} \cong \mathbb{Z}^6$, and the thirty-seam boundary has image exactly the even-sum sublattice D_6, with one residual parity coset. Under the pullback of the response-selected Gram metric, both integer modules embed densely in one abstract continuous three-dimensional Euclidean carrier. The sixty proper carrier maps act faithfully and isometrically on the rank-three range and extend to that completion, and cumulative records act by exact internal translations.*

The mechanism is spectral. The icosahedral adjacency has eigenvalues 5, $\sqrt{5}$, -1 and $-\sqrt{5}$ with multiplicities 1, 3, 5 and 3, so the repair mean M has eigenvalue $1 - (5 - \sqrt{5})/60$ on a three-dimensional eigenspace and smaller eigenvalues on everything else nonconstant. Repeated local averaging forgets every nonuniform pattern except a three-dimensional one, and that surviving pattern is the triplet of (2) carried by the port position vectors. The same number $(5 - \sqrt{5})/60$ reappears as the quantum spectral gap of the native repair channel (Section 3.3).

The completion is taken after slow-response normalization and the infinite-step limit; at finite step count the kernel has rank eleven on the centered space and rank six on the signed antipodal sector, so the order of limits matters. The metric is the response-selected Gram pullback, distinct from the usual six-dimensional lattice metric on D_6. Thirteen finite covariance and isometry table identities are discharged in Lean with `native_decide`, so those subreceipts trust the native compiler in addition to the kernel and are not kernel-only proofs. Displacement is a record difference and cumulative records translate the completion, so position is the carrier's own readback of its accumulated records. In the executable observer the same rank-three readout attaches to all joint histories of the interacting process, and a binary controller extension executes displacements spanning three independent directions [64]. Reading the completion as physical position consumes the realization map (M1) of Section 12.3 [1, 7]. The rank-three readout belongs to one carrier's

response. When carriers join into the refining spherical screen of Section 7.1, the continuum form of their repair history has spectral dimension two, the dimension of the screen, and bounded-energy families lose every within-carrier component, this band included, at the present full-port energy scaling; a three-dimensional bulk reading of the screen requires observation maps that the native energy does not supply.

6 The executable observer

Operational observers are continuing native processes that prepare, read, retain and act on records. The specification is executable, and its components run with retained receipts [64]. The companion construction paper states each component, its declared services and its certificates [63]; the following summary gives the parts used in this paper.

An observer that remembers and acts. The native observer executes two repair events with a finite tape. It records the first current, issues a forecast from that record, chooses the next seam from the record alone, records the result and halts. Exhaustive checks retain all 264 binary-source paths, and a 20,000-trial seeded execution reproduces the exact predictions. Record-controlled feedback gives final port-zero occupation 35/264, against 1/6 when the selector is disconnected or shuffled with the same action frequencies, so under the declared selector rule the response is caused by the retained record and not by the action statistics; the comparison checks the wiring of that rule. Feedback preserves the committed classical record while it can reduce coherence between different record values. Complete records of these short histories recover the source optimally, with entanglement fidelity $D^{-2}\sum_h w_h \operatorname{rank}(P_h)^2$ for the maximally mixed input, where D is the number of carrier configurations, h runs over the complete admitted histories, w_h is the product of their seam-selection and coin probabilities and P_h projects onto the inputs compatible with h; it equals 1301/2028 for the full-carrier two-event C observer and 3329/6084 for E under the same current-sign selector [67].

An electromagnetic detector driven by the closure coordinate. A second observer evaluates the coupling $\alpha(P)$ from the screen grain P and sets $g(P) = \sqrt{4\pi\alpha(P)}$. A prepared path qubit reads the first current through a calibrated loop exposure χ; the detector bit $y = \pm 1$ has probability $[1 + y\sin(g(P)j_1\chi)]/2$ and alone chooses the second seam, giving mean transferred charge $\mathbb{E}[g(P)j_2] = (5g(P)/264)\sin(g(P)\chi)$. All 528 weighted histories are retained and a 200,000-trial execution agrees with the analytic response. The closure coordinate thereby enters an actual response magnitude; the charge assignment, probe, exposure and controller are declared apparatus.

Fields measured by the observer. A scalar detector prepares a signed field intervention from the first native current, reads the field with a binary Weyl instrument and feeds the bit back, with inherited continuum bounds transferred to the complete history law. A source-compiled two-site field apparatus builds its entangled preparation from blank registers, evolves the field and reads it with a quantum pointer, all from emitted twelve-port control words; at the executed sixteen-level cutoff an exact rational interval places the ideal plus probability in $[0.51495, 0.51789]$, excluding a fair detector.

One interacting process. A common execution puts quantum electric flux, a charged fermion pair and two neutral scalar sites on one work register. Native source words implement their noncommuting evolution, the matter and scalar pointer instruments, and feedback controlled by a

committed matter bit; the final scalar bit controls the next native repair. A smooth extension retains four quantum reads on the same disturbed state, compiles the C and E record laws around it, and actively protects a logical qubit with record-controlled recovery. Its complete conditional native/work/reference outputs are retained, and Section 6.1 gives its certified responses.

Two observers, one law. Observer A reads ports $(0, 1)$ and observer B ports $(0, 5)$ of one carrier, so the two regions overlap at port zero. Each receives its initial bits and every incident local current, including zero currents, and its clock ticks whenever an attempted repair meets its region while remote repairs continue on the same source. After the first incident a native feedback repair is chosen from the reconstructed occupation of port zero. Both observers decode one continuing source, so their agreement at the shared port instantiates the overlap-agreement equation with identity transport; rechartings between observer charts are not exercised. A direct execution through the native endpoint branches and an independent recursion in a derived effective transfer representation agree exactly on all 5,768 output words, including timeouts and terminal populations; at every positive-weight prefix each observer's decoded fields obey native continuity, and the two values at the shared port agree [73]. The quantum extension checks the native remote resolvent on all 4,356 matrix units with coherent preparations and untouched references. A compiled version replaces its ideal preparation, intervention and pointer by pulses emitted from the source compiler: exact rational propagation of all 512 code addresses keeps every one of the 1,656 clock, timeout and abort outputs within 10^{-11} of the ideal history, at a cost of at most 51 pulses and five native repair attempts [74].

Local clocks. A local event clock counts the attempted repairs that meet its region, or the actual occupation changes at a port, and its stopped instrument sums the unresolved wait and keeps every finite timeout. Two such clocks on joined carriers share one continuing history with an observer-chosen field. C and E have the same completed clock law there and differ in the field: after three native attempts they predict field-plus probabilities $35/54$ and $14/27$ under the stated probe. Without feedback, over each two-tick cycle of one clock the other ticks twice on average with variance 16, so equal mean rates do not imply equal readings, and converting between the clocks through the joint event instrument preserves that whole count distribution [75]. A separate active clock lets each incident event, including a zero-current one, trigger one compiled modular step whose generator is the native three-state transfer, and closed histories derive a faithful thermal reference for that generator. Four movement-bit covariances on the complete attempt tape recover its two nontrivial frequencies and predict its coherent quadratures with no frequency fit [76]. Attempt counts, local event counts and modular parameters are distinct parameters, and none is identified with seconds.

Identifying the repair law. Nine source-executed scalar measurements identify all nine real coordinates of the record-law family and with them the coarse definite-current instrument on all 78 states, with arbitrary spectator and reference coherences. A collision test separates an admissible law that agrees with E on the whole binary shell. The calibrated instrument predicts complete two-event feedback histories through the same observer runtime, including quantum references and checkpoint continuation, and an explicit bound carries the probability uncertainty to their joint clock, current and interference records [77]. The probes are synthetic, so this identifies an instrument within the declared family and selects no law for nature.

From one carrier to a screen. One carrier suffices for the local results above; joined carriers test larger regions, distant interventions and refinement. Federations of 20, 80 and 320 twelve-port carriers preserve complete source records, untouched references and cross-carrier coherence probes, and local dependency bounds control the whole local quantum and record output under local feedback. On the actual 15,360- and 245,760-port screens, matched radius-nine initial restrictions give complete root-observer output distance below 3.82×10^{-4} at auxiliary per-seam parameter $1/4$, and below 6.656×10^{-9} after four local incident attempts with no auxiliary time coordinate [78]. On those matched screens a logical qubit with an untouched reference passes through two native incident ticks and local record-controlled feedback while the surrounding source keeps density $1/3$. A degree-two operator calculation retains every quantum output and timeout, and each full-screen output lies within 7.779×10^{-10} of the common 222-port computation, uniformly over references. C and E give identical classical records in this protocol and different reference coherences [79].

These executions are finite, and their apparatus, services, schedules, field action and preparations are declared; their time parameters are attempt and event counts. They discharge finite execution, continuation and comparison between observers under those contracts. A separate specified source clock shares a common calibration with complete field and repair operations. Autonomous unbounded observers, interacting spatial refinement, a physically validated clock/energy normalization and responsive gravity remain unestablished (Section 12).

6.1 Measurement inside one interacting process

OPH places measurement at the transition from a private repairable record to a protected public normal form: a record is public when it has survived every authorized comparison and no repair can change it, and event projectors, weights and updates act on the completed algebra-state surface of Section 4.3. The executable observer puts this account inside one interacting quantum process [63].

Quantum electric flux, a charged fermion pair and two neutral scalar sites share one work register, with operator Gauss law and source-executed instruments. The sharp protocol retains 1,056 native/quantum histories. At $d = 2$ and 512 temporal steps its certified feedback contrast is $[-0.033297300667839, -0.030025660061775]$; for zero native preparation current the actual infinite-coordinate half-line instrument gives $[-0.024045670143, -0.024043450955]$, an infinite-occupation enclosure that includes the free vacuum and unbounded interacting evolution of the same declared action [68].

Outcome accuracy does not certify the output state. For that zero-current preparation, the complete $d = 2$ sharp output has trace distance about 0.455 from the infinite history; every output in the $d = 256$ occupation code has error above 0.122. For the vacuum control the best code error scales as $d^{-1/4}$ [69]. A separate smooth Ramsey instrument preserves finite number moments. A four-read continuation carries its disturbed state through two smooth reads, feedback and a later gauge probe, retaining sixteen work branches. Stagewise moment and projected-Duhamel estimates give complete $O(1/d)$ error at fixed protocol with $n(d) = d^2$ temporal steps; executed $d = 2, 4$ occupation bounds remain vacuous. This covers the later probe, not post-sharp clocks or physical clock calibration [70].

Compiled C/E native repairs surround that same work history, acting on arbitrary joint inputs of the native, work and reference registers. They retain full conditional quantum outputs, aborts and padding. If r_2 is the second smooth bit, $K_r^{j(h)}$ the work branch, and $e(0) = (0, 5)$, $e(1) = (0, 1)$, the joint Kraus map is

$$M_{h,r,s} = (L_s^{e(r_2)} L_h^{(0,1)}) \otimes K_r^{j(h)}.$$

Causal summation proves completeness. A retained native phase witness has C contrast $(1+q)/4$, $q = \Pr(r_2 = 0 \mid j = 0)$, and E contrast zero; source execution gives approximately 0.3492 and 0.3514 for C at $d = 2, 4$ [71].

Active memory uses the existing four-configuration qubit V and compiled record-controlled recovery $R_h L_h V = \sqrt{p_h} V$. Correcting after each native event, with final seam $(2, 3)$ for $r_2 = 0$ and $(0, 1)$ otherwise, preserves the logical/reference marginal even for entangled work inputs. For independent logical/work preparations, the matched passive complete plus/minus output distance is $1 - \bar{q}$, $\bar{q} = (q_{-1} + 2q_0 + q_1)/4$, $q_j = \Pr(r_2 = 0 \mid j)$, versus one with correction; a terminal phase read consumes that coherence [72]. This protects an encoded qubit under specified controls, not arbitrary lost native coherence.

7 Fields, clocks and causal order

The field constructions distinguish a native screen limit, a chosen radial field service, and a three-dimensional fermion refinement. Their common physical interpretation requires compatible state, current and clock maps; sharing an operator name does not establish those maps.

7.1 The native screen operator and a conditional spacetime field

On the geodesic screen of level l, let $V_l = 240 \cdot 4^l$ and prepare the uniform canonical state with total load $V_l/3$. Independent uniform seam selection and fair native repair define one complete occupation history. Write $\chi_l = q_l(V_l - q_l)/[V_l(V_l - 1)]$ for $q_l = V_l/3$.

Proposition 7.1 (Native continuum operator)**.** *Along a common refinement subsequence, the native cell Green operators converge in norm to a compact positive injective operator G on the mean-zero observation space. After restoring the constant mode, its inverse A generates a regular strongly local conservative form, with domain H^1, and*

$$n/49920 \le \lambda_n(A) \le 3n/2.$$

The complete density readouts

$$Y_l(\tau; f) = \frac{\sum_p f_l(p)[n_{\lfloor 1260\, 16^l \tau \rfloor}(p) - 1/3]}{\sqrt{V_l \chi_l}}$$

converge jointly at finite sets of times to a stationary reflection-positive Gaussian field with covariance $\langle f, e^{-|\tau - u|A} g \rangle$. The admitted coherent continuation converges strongly to $\cos(t\sqrt{A})$. With the constant mode restored,

$$\|\mathbf{1}_F e^{-\tau(A \oplus 0)} \mathbf{1}_E\| \le e^{-d(E,F)^2/(2\tau)},$$

and the cosine propagator vanishes for $|t| < \sqrt{2}\, d(E, F)$.

Proof sketch. The native cell Poincaré bound $\|(I - P_k)u\|^2 \le 624\, 4^{-k} \langle u, B_l u \rangle$ gives uniform compactness of the Green operators and norm convergence. Mosco convergence passes their local Markov forms to the limit; min–max estimates give the displayed spectrum. Conditional finite-population limits apply to the actual identity-or-swap repairs. The Gaussian heat bound gives finite propagation of the admitted cosine family [90, 91, 92, 80, 81]. □

Thus one source operator controls static covariance, relaxation and a chosen coherent wave service. Its spectral dimension is two. The $\lambda_n \asymp n$ count excludes identifying it, up to scale

and bounded mass shift, with a three-dimensional elliptic bulk operator or a first-order boundary operator. A physical bulk interpretation needs an additional radial observation map. Passive repair supplies no wave sector; its attempt count and the coherent wave parameter require clock calibration. The native stationary reference supplies no physical quantum vacuum or universal area-entropy coefficient [63, 82].

A specified radial interval and field service give the product covariance $(-\partial_r^2 + A \oplus 0)^{-1}$. If its hyperbolic coefficient density is $C = \mu\ \mathrm{diag}(-1, a)$, the effective metric is

$$g = \sqrt{-\det C}\, C^{-1} = \mu\sqrt{\det a}\ \mathrm{diag}(-1, a^{-1}).$$

The spatial coefficients are read from the source; the radial architecture, hyperbolic service and preparation are explicit design inputs.

Proposition 7.2 (Smooth effective field descriptions)**.** *Smoothing the native angular coefficients while retaining the radial Dirichlet interval gives norm-resolvent convergence, with gap $(\pi/L)^2$. For every fixed finite admissible menu of observer readouts on bounded wave times and every $\varepsilon > 0$, a smooth static $(1 + 3)$-dimensional Lorentzian metric carries a free scalar field whose complete predictions agree within ε, in addition to the stated source and clock errors.*

The proof transports the changing measures into one Hilbert space and combines compact Mosco convergence with continuity of the covariance, response and finite-history operators [63]. It proves a common finite-menu approximation, without selecting a radial architecture, providing a smoothing rate, or producing a responsive metric. The three-dimensional fermion limits in Section 9 have their own source and refinement hypotheses.

7.2 A causal precursor from records

The oriented conformal spherical support carries the classical isomorphisms

$$\mathrm{Conf}^+(S^2) \ \cong\ \mathrm{PSL}(2,\mathbb{C}) \ \cong\ \mathrm{SO}^+(3,1), \qquad H^3 \ \cong\ \mathrm{SO}^+(3,1)/\mathrm{SO}(3), \tag{7}$$

which identify the three-dimensional space of future unit timelike directions [49]. Authenticated records separately supply events and their order. Every quotient-visible register version carries the identifier of its last semantic writer, each commit certifies the versions its acceptance depends on, and the informational order is the transitive closure of authenticated read-from parenthood, an authenticated analogue of happened-before [39]. As an abstract finite poset it satisfies the order-theoretic causal-set axioms, and every finite causal set is the generated order of some authenticated log [1].

Theorem 7.3 (Source-derived finite (1+3) causal precursor)**.** *Let $\mathcal{L}$ be an authenticated finite semantic-event log with generated strict order $\prec_{\mathcal{L}}$ and canonical source height $h_{\mathcal{L}}$, zero on roots and one plus the maximum parent height otherwise. Then $h_{\mathcal{L}}$ strictly increases along $\prec_{\mathcal{L}}$ and equals the maximum length of an authenticated parent chain ending at each event. Let V_{src} be the rank-three source Gram quotient with positive form g_{src}, and put $W_{\mathrm{src}} = \mathbb{R} \oplus V_{\mathrm{src}}$ with $Q(t, x) = t^2 - g_{\mathrm{src}}(x, x)$.*

(F1) W_{src} has dimension $1 + 3 = 4$ and Q has one positive and three negative directions;

(F2) source-unit directions, S^2 and future-null rays are exactly equivalent;

(F3) a placement $F(e) = (\tau h_{\mathcal{L}}(e), x(e))$ sends every generated precedence into the future cone whenever each parent edge obeys $g_{\mathrm{src}}(x(f) - x(e), x(f) - x(e)) \le \tau^2 (h_{\mathcal{L}}(f) - h_{\mathcal{L}}(e))^2$;

(F4) every finite log admits an injective placement of this form that sends generated ancestry into the future cone and makes distinct same-height events spacelike; and

(F5) for a placement obeying the edge condition of (F3), if the spatial readback separates each equal-height layer and every unrelated pair with $h_{\mathcal{L}}(e) < h_{\mathcal{L}}(f)$ is spacelike, then $e \preceq_{\mathcal{L}} f$ holds exactly when $F(f) - F(e)$ is future causal, so F is an exact order embedding preserving every causal interval.

Proof. Dimension is additive and positivity of g_{src} gives the inertia; induction over parents bounds every ending chain by the height and constructs an attaining chain; the edge inequality and transitivity of cone and ancestry give (F3); an enumeration along one axis with $\tau = n + 1$, n the number of events, gives (F4); and the two geometric checks exclude every unsupported cone comparison in (F5). The finite statements are formalized [1, 9]. □

Read informally, the theorem combines two things observers have. Who read whose write is a partial order, and the longest chain of reads ending at an event gives it a height. The rank-three carrier gives three spatial directions and, with a real axis, a light cone. An event placement that respects both turns the provenance order into a causal order in a four-dimensional Lorentzian carrier.

The first two clauses construct the carrier and signature from an independent real axis and the rank-three source carrier, without assuming a rank-four chart; source height enters only through placements, and the scaled height is no operational clock. A physical manifold requires a source-causal continuum certificate: one refinement family supplying physical event identification, source-selected order and cone embeddings, refinement maps, a calibrated count density, agreement across independent dimension and manifoldlikeness tests against Poisson-sprinkled controls, and convergence to a distinguishing time-oriented Lorentzian geometry [30, 31, 32].

Proposition 7.4 (Composition of the continuum certificate)**.** *For a source-causal continuum certificate, causal order fixes the conformal Lorentzian geometry of the admitted distinguishing limit and the calibrated count measure fixes its conformal scale [37, 38]; the local model is $(1 + 3)$, so a limit satisfying the dimension and topology clauses of the certificate is an effective four-dimensional time-oriented Lorentzian event manifold. If on the same family a causal-set d'Alembertian or scalar-curvature estimator converges as an independent curvature diagnostic [35, 36], an explicit tensor-curvature reconstruction converges to the Einstein tensor or the small-ball null-balance route identifies the limiting geometry tensor with G_{ab}, and stress and Ward–Bianchi data converge with calibrated coupling, then the smooth limit is $G_{ab} + \Lambda g_{ab} = 8\pi G\, T_{ab}$.*

A source-record population realizes the flat case.

Theorem 7.5 (A source-record causal and count limit)**.** *In the rank-three conservative repair carrier, let $\varphi = (1 + \sqrt{5})/2$, $L = 2/\sqrt{\varphi + 2}$ and $q = F_n$ the n-th Fibonacci number, $n \geq 4$. The population $S_q = L\{b\varphi - \lfloor b\varphi \rfloor : 0 \leq b < q\}^3 \subset [0, L]^3$ consists of q^3 distinct conservative source-record images. Retain it at each layer and let every next-layer event read all preceding-layer sites within $a_q = L/\sqrt{q}$, with model duration $\Delta_q = a_q/c$ for a declared speed c. As $q \to \infty$ the order converges away from null separation to the speed-c causal order, and with $\rho_q = q^3/(L^3\Delta_q)$, for fixed interior timelike diamonds,*

$$\frac{|I_q|}{\rho_q} \longrightarrow \mathrm{Vol}(I), \qquad \frac{2C_q}{|I_q|(|I_q| - 1)} \longrightarrow \frac{1}{10},$$

where C_q counts strict causally ordered pairs; for vertical duration T, $\mathrm{Vol}(I) = \pi c^3 T^4/24$.

Corollary 7.6 (Proper-time ratios from record counts)**.** *If I_q, J_q approach interior timelike diamonds of proper durations $\tau_I, \tau_J > 0$, the observer readout $R_q(I;J) = (|I_q|/|J_q|)^{1/4} \to \tau_I/\tau_J$. Within positive volume-only readouts additive on collinear inertial intervals, the fourth root is unique up to units.*

At finite regulator the population is produced by the carriers themselves: each holds its record as port loads, reads its position through the rank-three response and decides its neighbors from that readback, and the precedence generated from their authenticated reads equals the layered order through $q = 89$, that is 704,969 carriers reading one another over ten rounds. On the largest interior diamond the inverted Myrheim–Meyer dimension reads 2.83, 4.38, 3.55, 3.84 and 3.92 at $q = 13, 21, 34, 55, 89$ [33, 34]; the outermost central diamonds, which the cube clips and which are therefore not interior readings, read 4.15, 3.61, 4.10, 4.09 and 4.07, and the readings at $q = 5$ and $q = 8$ lie below the regime of the theorem. The ordering fraction of the central diamond reads 0.088, 0.138, 0.092, 0.093 and 0.094 against the proved flat value 1/10, and the count clock reads 1.71, 2.09, 1.91 and 2.05 at $q = 13, 34, 55, 89$ against the model-time ratio 2. Covariance requirements select the cone: a closed convex pointed displacement cone preserved by the irreducible icosahedral rotations and by boosts along one axis is the future or past cone of $t^2 - \|x\|^2$. The read law and model clock are declared, and the ordinal height of a history with self-waits is no proper-time instrument [9].

The executable carrier has its own finite causal bound. For a fixed seam schedule and shared local coins, dependence crosses at most one seam per ordered repair event, and retained signal witnesses attain the bound; this is a speed in seams per native event. Remote repairs can carry a difference to a receiver before its first local tick, so the bound is not one seam per receiver tick [83, 64]. Complete endpoint records give a forward-cone placement in the registered response geometry, and a four-event native execution has a Lean instance proving that implication and the failure of its converse [1].

8 Gauge structure, particle operators and the pixel scale

The finite theorem fixes a current algebra under explicit response premises. A chosen carrier, memory law and field service then realize its currents and mass operators. These are different steps of one construction: the Lie type constrains admissible interactions, while the full executable specification fixes their coefficients. Recovering or choosing that specification is allowed; its agreement with physical particle records is an additional requirement.

8.1 The finite gauge theorem and matter representation

Let A be the icosahedral adjacency matrix and J its antipodal involution. Exact incidence gives

$$10J = A^3 - 4A^2 - 5A + 10I, \qquad P_{12} = \mathbf{1} \oplus \mathbf{5} \oplus \mathbf{3} \oplus \mathbf{3}'. \tag{8}$$

The even port coordinates carry $\mathbf{1} \oplus \mathbf{5}$; the odd coordinates carry the two triplets. The response premise of Section 2.2 requires an injective derivative $D : P_{12} \to \mathfrak{u}(H)$, a commutator-closed image with positive trace pairing, and internal implementation of every proper carrier rotation. Internal implementation means that its projective implementer factors into the connected response group and a pointwise centralizer. This is the response-endogeneity clause identified as A2 in the library; it is not implied by the port count alone [94, 11].

Theorem 8.1 (Gauge Lie-type forcing)**.**

Under these complete-response and internal-implementation premises,

$$\mathfrak{g} = D(P_{12}) \cong \mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3).$$

Its center is the uniform port line. The three-dimensional simple ideal carries one triplet, and the eight-dimensional ideal carries the other triplet and the five-dimensional band.

Proof. The trace pairing makes the twelve-dimensional image compact. Internal implementation makes the A_5 action inner, so it fixes the center pointwise. The transitive port action has a one-dimensional fixed space, hence the center has dimension at most one. With zero center, the only compact semisimple type of dimension twelve is $\mathfrak{su}(2)^4$. Inner automorphisms preserve its simple ideals, each of which has either zero or three A_5-fixed directions, contradicting the one-dimensional fixed space. The center therefore has dimension one; the unique compact semisimple type of dimension eleven is $\mathfrak{su}(2) \oplus \mathfrak{su}(3)$ [60]. □

The premises have an explicit finite matrix realization. Six antipodal axis vectors form a matrix U over $\mathbb{Q}(\sqrt{5})$; $\sigma(U)$ is its Galois conjugate. For even coordinates b, odd coordinates d, and $c = \frac{1}{6}\sum_i b_i$, put $\Phi_0(b - c\mathbf{1}) = \sum_i (b_i - c) u_i u_i^{\mathsf{T}}$. For nonzero real band coefficients,

$$K(f) = \operatorname{diag}\Big(\lambda_3 \widehat{Ud} + i[\lambda_1 c I_3 + \lambda_5 \Phi_0(b - c\mathbf{1})], \lambda_{3'} \widehat{\sigma(U)d}\Big), \qquad \widehat{x}y = x \times y. \tag{9}$$

Proposition 8.2 (Finite response realization)**.** *The map K is injective and equivariant, its image is commutator-closed, and proper carrier rotations act internally. Its pulled-back bracket has*

$$(P_{12}, [\ ,\ ]_K) \cong \mathfrak{u}(3) \oplus \mathfrak{so}(3). \tag{10}$$

The six axis projectors span the symmetric matrices, while the triplets span the skew matrices in the two blocks. These rank and equivariance identities are exact over the full sixty-element rotation group. The representation and nonzero coefficients are design inputs. A zero bracket on the same port module violates the internal-implementation premise; it is not excluded by geometry alone. The gauge specialist gives the complete finite certificate and the boundary between analytic classification and machine-checked algebra [11].

Theorem 8.3 (Finite matter image)**.** *For the supplied trace-balanced module $V = \mathbb{C}^3_{-1/3} \oplus \mathbb{C}^2_{1/2}$,*

$$\Lambda^2 V \oplus \Lambda^4 V = Q \oplus u^c \oplus e^c \oplus d^c \oplus L, \tag{11}$$

where $Q = (3,2)_{1/6}$, $u^c = (\bar{3},1)_{-2/3}$, $e^c = (1,1)_1$, $d^c = (\bar{3},1)_{1/3}$ and $L = (1,2)_{-1/2}$. All perturbative gauge and mixed gravitational anomalies vanish, the number of weak doublets is even, and the faithful image is

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}. \tag{12}$$

Among the 1,024 *subsets of the ten nontrivial irreducible exterior summands, nonemptiness, chirality and the four anomaly-trace conditions select exactly this fifteen-state mask and its charge conjugate.*

The exterior branching, anomaly traces and center kernel are finite calculations [11]. The supplied module is the familiar $\mathbf{10} \oplus \bar{\mathbf{5}}$ branching; this result does not require a physical SU(5) gauge field. It fixes neither three families nor their mass matrices.

8.2 The action and its coefficient realization

Three declared family copies, a Higgs doublet, canonical kinetic terms, local Lorentz covariance and the dimension-at-most-four interaction grammar produce the conditional Standard Model action. The finite current equations fix its invariant Yukawa tensors; their three family matrices are free until the memory and feedback specification is given. Their singular values set the tree mass coefficients $m_f = v\, s_f(Y_f)/\sqrt{2}$. This is an exact conditional action statement, with physical pole matching and continuum control as separate requirements [12].

The retained pixel-dependent charged-lepton prescription gives coefficients $t_j(P) = m_j(P)/v(P)$. Three native port projectors P_2, P_3, P_{10} and a complete reset-memory process realize a positive matrix $M(P)$ satisfying

$$\operatorname{spec} M(P) = \{t_1(P), t_2(P), t_3(P)\}, \qquad M(P) = g Z_C.$$

Here Z_C is the unconditional C-duty coefficient, whose trace is the C-event probability p, and $g = \sum_j t_j/p$. Independent left and right probe durations give two admissible controller solutions. Disabled records are retained. The mass-law architecture has target-informed provenance; its execution reads no measured lepton masses. This is a realization of that specified law, without a claim of independent empirical prediction. The same-current, canonical-kinetic family freedom is $U(3)^5$, which cannot change invariant mass ratios by relabeling [12].

The pixel equation must use those same coefficients in its response. Section 11.2 states this coupling explicitly. A loop built with a different mass law or field census is not the fixed point of this chosen observer model.

8.3 Localized modes, topology and physical particles

A massive particle interpretation requires a localized excitation with persistent charges, a finite total energy, consistent exchange and decay rules, and a moving-state dispersion in the common clock. A topological invariant can constrain such states; it does not by itself fix their masses or identify their representation with an observed species.

For the radial two-spin source, an exact two-column kernel isometry W satisfies $hW = 0$. With $C = D_{\text{stagger}} i\sigma_2$ and $C\overline{W} = W i\sigma_2$, extend the carrier by the native matter module and its three families. The local pairing $\Delta = C \otimes \overline{K}$ has an exact Nambu inclusion

$$E = \operatorname{diag}(W \otimes I_{45}, \overline{W} \otimes I_{45}), \qquad BE = E B_{\text{core}}.$$

All twelve native currents intertwine through this same inclusion. Choosing $Y_e = \sqrt{2} M(P)$ puts the retained lepton coefficients on these same localized states. Every complete record-driven pairing pulse preserves the inclusion. The bulk bands are additive $\lambda(h) \pm s_j(K)$, which supplies no relativistic mass shell. Three family copies are input, and creation of a bound fermion does not create the prescribed winding texture [12].

A compact alternative has two cores with shell degrees $+1$ and -1 and a constant exterior. Its spin energy and fermion vacuum-energy difference have bounds independent of exterior volume. For texture amplitude $M \geq 8$ there are exactly four localized modes, with explicit exponentially small outer-boundary errors. A complete rational certificate excludes zero from the four levels at the displayed separation-one source. Writing $A = W^\dagger h W$, the exact pairing inclusion retains $A \neq 0$; its bound energies shift the family coefficients. This compact band has four spatial columns, whereas the radial kernel has two. Neither its multiplicity nor its finite pair energy is an identification of particle generations or constituent rest masses [12].

Distinct protected windings also cannot label arbitrary generations: additive charges must respect the actual mixing graph and Majorana vertices. The particle specialist gives these constraints, the complete finite spectra and their current maps. Autonomous defect formation, physical spin and statistics, pole masses and compatible massless sectors are not supplied by the finite inclusions [12].

9 Interacting finite matter and controlled field limits

The executable specification determines complete instruments, memory updates and field services. Coherent ordering memory realizes the $\mathbb{Z}/2$ exchange-history charge of the native two-load configuration complex; its colored extension has fermionic exchange. Complete field factors then act on the full occupation algebra with a common calibrated clock and retained records. These finite constructions support actual state and operation maps, rather than a match between isolated eigenvalue lists [63, 12].

9.1 A nonquadratic dynamics on the localized algebra

On the radial ninety-mode localized algebra of Section 8.3, add to every record Hamiltonian the chosen interaction

$$V = \lambda(N_W - 45)^2, \qquad \lambda \geq 0.$$

It is parity even and commutes with every native internal current. The localized and complementary CAR factors reduce every record Hamiltonian; therefore the same state inclusion and the full record process preserve the interacting localized algebra exactly. For C-event probability p and feedback gain g, every localized record Hamiltonian has norm at most

$$M_W = 45(1+g) + 2025\lambda.$$

Complete-memory contraction gives, for n pulses in time t and arbitrary initial memory correlations and finite untouched references,

$$\|\rho_{W,n}(t) - e^{-it\bar{H}_W}\rho_W(0)e^{it\bar{H}_W}\|_1 \leq \frac{4M_W t}{n(1-p)} + \left(\frac{8}{1-p} + 4\right)\frac{M_W^2 t^2}{n}.$$

This $O(1/n)$ bound, denoted δ_n, compares the unread final field and reference after summing every native outcome. The complete final-history comparison obeys the separate bound

$$\|\Omega_n - \Omega_n^{\mathrm{ideal}}\|_1 \leq \min\{2, \sqrt{2\delta_n}\} = O(n^{-1/2}).$$

For each initial classical memory label m, the ideal keeps the actual conditional history law $p(\omega \mid m)$ and applies the mean unitary to the corresponding initial field/reference state ρ_m. It therefore retains initial memory correlations. The bound follows from conditional unitary evolution, the pure-state trace-distance identity and convexity; it supplies no uniform error after conditioning on a rare outcome. Both comparisons are uniform in the containing radial cube because the observed algebra has fixed dimension. The quartic is exponentially approximable by finite-region operators, but it is not a strictly onsite interaction or a translation-invariant many-defect field theory [12].

The interaction changes the spectrum. An exact sixteen-state reducing slice of the mean Hamiltonian has

$$H_4 = m(a_1^\dagger a_2^\dagger + a_2 a_1 + a_3^\dagger a_4^\dagger + a_4 a_3) + \lambda(N_4 - 2)^2.$$

Its single-fermion spectral poles and addition-plus-removal residues are

$$\omega_\pm = 2\sqrt{\lambda^2+m^2} - \lambda \pm m, \qquad w_\pm = \frac{1 \mp m/\sqrt{\lambda^2+m^2}}{2}.$$

A nonzero cross-pair density cumulant proves a non-Gaussian state. This slice is not the full physical vacuum, and individual memory records need not preserve it. The compact four-column band has $A \neq 0$ and a different mode count, so these radial slice formulas do not transfer to it unchanged.

The same finite interacting Hamiltonian admits a faithful Gibbs reference by a declared preparation service. It obeys $-\log\rho = \beta H + \log Z$ and the noncommuting response

$$\rho' = -\beta \int_0^1 \rho^s (X - \langle X\rangle)\rho^{1-s}\, ds, \quad F'' = -\beta\, C_\rho(X,X) \leq 0, \quad S' = \beta(E' - \langle X\rangle).$$

Here C_ρ is the positive logarithmic-mean covariance. These identities join interaction, energy and entropy on one finite algebra. A coupling perturbation X is not thereby identified with geometric stress, and Gibbs stationarity is not fixed-volume geometric entropy stationarity.

9.2 What survives spatial refinement

At fixed lattice spacing the three-dimensional Weyl/contact source has an infinite-volume positive-energy ground, a nonzero connected four-point function, and a time-reflection-positive Euclidean hierarchy. These are interacting lattice statements. With $c = v/a$ and the centered clock $H - (U/2)N$, uniform spectral estimates show that the continuum is the free massless field in all eight Weyl valleys throughout

$$0 \leq U, \qquad U = o(c^{2/7}), \qquad c = v/a \longrightarrow \infty.$$

The full valley census is retained; the result supplies neither a chiral Standard Model continuum nor massive physical species [63].

A distinct scaling, $c = J|E|^4$ and $U = J|E|^8$, produces interacting Heisenberg dynamics at fixed spacing. Matched infrared bounds give an ordered ground with $m_s > 1/1000$ and a regular canonical fluctuation pair with continuous translations, finite propagation, and positive-energy weight at arbitrarily small momenta. Its dynamics is gapless and nonstatic; a sharp relativistic mass shell and a surviving interacting relativistic limit are not established. Native finite-history approximations have explicit locality and control errors, without an effective finite-size modulus for preparing the infinite selected ground [63].

For the free vacuum at $U = 0$, conserved Wick stress and geometric ball modular flow give the charge $(2\pi/v)\int (R^2 - r^2)\mathcal{E}/(2R)\, d^3x$. Nine independent shell preparations approach the same leading stress response with derivative control. Their relative entropy has an explicit sixth-order coefficient and an $o(R^4)$ nonlinear remainder in the stated lattice/small-ball regime. These results constrain a gravity construction; responsive area and one common interacting state satisfying the geometric stationarity premise are separate requirements [9].

10 The Einstein equation from a realized null balance

The gravitational hypothesis identifies Einstein dynamics with the coarse response of the same process.

Theorem 10.1 (Null tomography and metric ambiguity)**.** *Let V be four-dimensional with Lorentz form η of signature $(-,+,+,+)$, the opposite convention from Q in Section 7.2; nothing below depends on the choice. If a symmetric form X satisfies $X(k,k)=0$ for every null vector k, then $X=\phi\eta$. Null-null values therefore determine a symmetric tensor modulo the metric line, and nine explicit null directions, the six axis rays $(1,\pm e_i)$ and $(1,s,s,\pm s)$, $(1,s,-s,s)$ with $s=\sqrt{3}/3$, reconstruct the quotient $\mathrm{Sym}^2(V^*)/\mathbb{R}\eta$ with design determinant $8192/27$ in the certificate's basis and a decoder obeying $\|x\|_\infty \le (2+\sqrt{3})\|q\|_\infty$.*

Proof. On future null rays $(1,n)$ the identity $X_{00}+2X_{0i}n_i+X_{ij}n_in_j=0$ on the unit sphere forces $X_{0i}=0$, $X_{ij}=\phi\delta_{ij}$ and $X_{00}=-\phi$. The determinant and decoder bound are exact calculations, and the design, injectivity and ambiguity are formalized in Lean [1, 9]. □

Theorem 10.2 (Tensor completion of a realized null balance)**.** *Let (M,g) be a connected, time-oriented, C^3 Lorentzian four-manifold and T_{ab} a symmetric C^1 tensor with $\nabla^a T_{ab}=0$. Suppose a common realized tower supplies $\kappa=8\pi G_N>0$ with*

$$G^{(g)}_{ab}k^ak^b=\kappa T_{ab}k^ak^b \tag{13}$$

for every null vector k at every point. Then there is a constant Λ with

$$G^{(g)}_{ab}+\Lambda g_{ab}=8\pi G_N T_{ab}, \tag{14}$$

and a reference event at which the difference equals $-\Lambda_0 g_{ab}$ fixes $\Lambda=\Lambda_0$.

Proof. Theorem 10.1 applied pointwise to $G^{(g)}-\kappa T$ gives $G^{(g)}_{ab}-\kappa T_{ab}=\phi g_{ab}$. The contracted Bianchi identity, the Ward premise and metric compatibility give $\nabla_b\phi=0$, connectedness makes ϕ constant, and $\Lambda=-\phi$. □

The structure of the argument is simple. Light rays cannot see a stress term proportional to the metric, so balance between curvature and stress on every null direction fixes Einstein's equation up to such a term, and the Bianchi identity with stress conservation makes its coefficient constant. That constant is the cosmological constant. The coefficients of entanglement-equilibrium arguments [56] enter through one machine-checked identity: scalars with $\delta S_{\mathrm{bulk}}=\frac{8\pi^2\ell^4}{15}t$, $\delta A=-\frac{4\pi\ell^4}{15}f$ and $\delta S_{\mathrm{bulk}}+\delta A/(4G_N)=0$ obey $f=8\pi G_N t$. For OPH to supply (13), one common refinement tower must realize (G1) the effective manifold of Proposition 7.4; (G2) geometrically normalized cap modular flow whose directional charges define one stress tensor; (G3) its Ward identity, independently of that reconstruction; (G4) the edge-center entropy split, the normalization $\delta\langle Z_L\rangle=\delta A/(4G_N)$, the modular first law and generalized-entropy stationarity; (G5) a small-ball family with controlled remainders and enough observer directions; and (G6) universal coupling, a vacuum reference and a scale calibration fixing $\kappa=8\pi G_N$. The finite layer contains null tomography, the edge-entropy split and the coefficient identity, and the machine-checked composition proves the tensor step from an explicit null-balance premise; the bridge between them is (M4).

Section 9 supplies sector-relative modular and Ward inputs to (G2)–(G3) for the proved free Weyl sector, and the matter first-law and small-ball parts of (G4)–(G5). Nine explicit shell perturbations span traceless stress variations; finite boosted-ball responses have full rank for $0<R/L\le 2^{-28}$. At exactly $U=0$, compact lattice states and quadratic detectors recover the leading stress response in a joint spacing/small-ball limit. On each fixed ball, the actual finite-density modular flows also converge on all fixed probes, with a generator graph core and convergent selected relative entropies. The proofs retain all eight valleys and do not assume a uniform covariance gap; see `model/lattice_modular_limit/` and `model/lattice_weyl_response/`. The join to native

event geometry, the edge/area law, generalized-entropy stationarity and universal coupling with a calibrated G_N remain open. Neither a conserved flat-space stress nor its modular kernel alone implies curvature response.

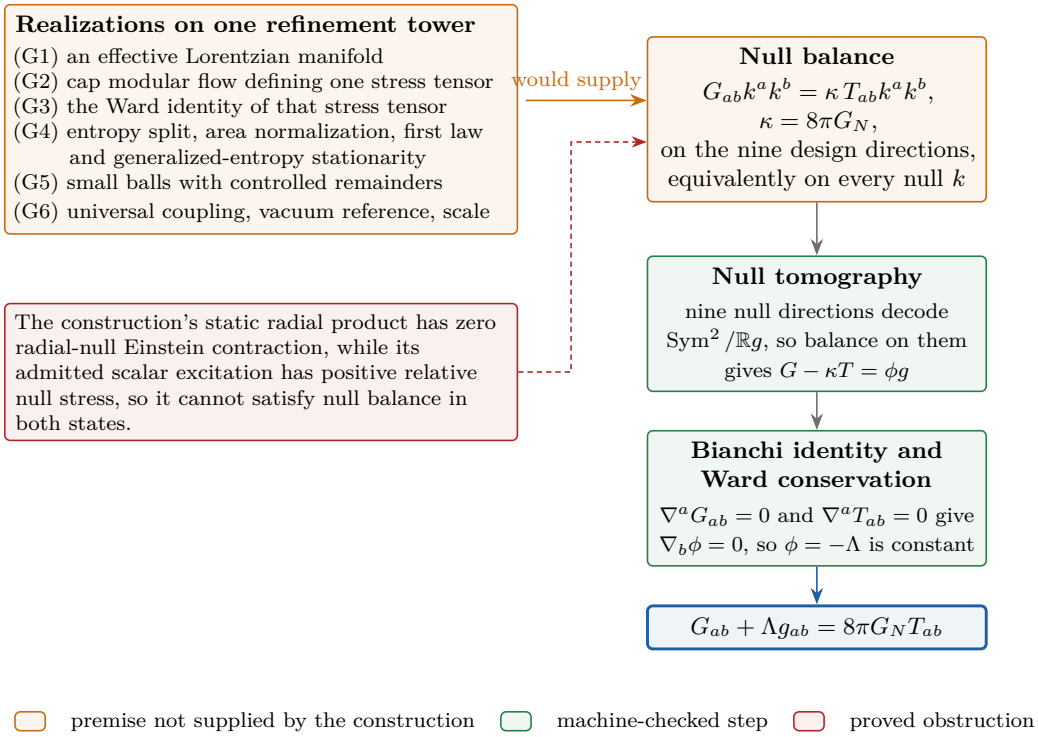


Figure 3: The Einstein equation as tensor completion of a realized null balance (Theorem 10.2). The tensor step is machine-checked from an explicit null-balance premise. Realizations (G1) to (G6) on one refinement tower would supply that premise, and no complete realization of them on one common refinement tower is established; the tested static radial product fails the balance for its own scalar excitation.

11 Cosmology and the closure coordinates

The cosmological hypothesis asks the same process to fix the vacuum, the dark sector and the primordial response. In this paper the layer consists of conditional branches and diagnostics of declared maps.

11.1 The dark sector: non-luminous charge, the deep galaxy law, and the vacuum coordinate

Under supplied universal-coupling, physical-stress and source-localization premises, sourcing gravity by modular charge classifies any geometric excess over the luminous term as non-luminous modular charge, with anomalous modular energy on overlap collars as one candidate carrier [15]. Conditional on two named premises the deep-regime profile is characterized rather than supplied: an enclosed-mass law that is degree-one scale covariant and composes independent sources in quadrature carries a unique constant $a_0 > 0$ with $M_A(r) = r\sqrt{M_b a_0/G}$, which forces flat rotation curves and gives $v^4 = GM_b a_0$, the baryonic Tully–Fisher relation, each premise load-bearing by an explicit

counterexample. A per-cut model gives $a_0 = Gn^2c$, with only the product n^2c identifiable, and the value of a_0 is fixed by no source law. Reading the dark-energy density as $\rho_{\rm DE} = \kappa/N$ at constant κ gives $w = -1 + \frac{1}{3}\,\mathrm{d}\ln N/\mathrm{d}\ln a$, so fixed capacity gives the exact point $(w_0, w_a) = (-1, 0)$ and nondecreasing capacity gives $w \geq -1$; exclusion of that point fails the fixed branch, while agreement supports nothing, since it is also the ΛCDM null [16].

11.2 Pixel closure and the common particle response

The dimensionless pixel coordinate is $P = a_{\rm cell}/\ell_P^2$. Its self-reference equation relates it to a response computed from the chosen source specification,

$$P = \varphi + \frac{\sqrt{\pi}}{A(P)}, \qquad \varphi = \frac{1+\sqrt{5}}{2},$$

where A is the declared inverse electromagnetic-coupling readout. A fixed-point proof determines a root of that map. Identifying its readout with the physical Thomson coupling requires the current response of the same action, vacuum, spectrum and clock.

One explicit conditional map joins the native matter census to the charged-lepton law of Section 8. For unified coupling u, energy unit E_* and scale coefficient β, take

$$\frac{M_U}{E_*} = e^{-2\pi}P^{1/6}, \qquad \frac{v}{E_*} = P^{-1/2}e^{-2\pi/(\beta u)}, \qquad \alpha_i^{-1} = u^{-1} + \frac{b_i}{2\pi}\log\frac{M_U}{m_Z}.$$

The infinite heat-sum equation, with certified tail bounds, supplies u; the same neutral-Higgs relation supplies m_Z/v. Exact charge and Casimir traces on the fifteen native matter rows, with three families and one complex Higgs doublet, give

$$(b_1, b_2, b_3) = (41/10, -19/6, -7)$$

when inserted into the declared four-dimensional one-loop functional. The representation traces are finite consequences of the gauge module; the loop functional and scale law are additional physical modeling inputs. This is the intended surviving chiral census; applying it to the eight-valley lattice requires a proved mode-removal or threshold map [11, 10, 12].

For every charged lepton and all six quarks, the same coefficient masses enter the response kernel

$$K_f = \frac{2N_cQ_f^2}{\pi}\int_0^1 x(1-x)\log\left(1 + \frac{m_Z^2}{m_f^2}x(1-x)\right)dx,$$

$$A = \alpha_2^{-1} + \frac{5}{3}\alpha_1^{-1} + \sum_\ell K_\ell + \left(1 - \frac{3\alpha_3}{\pi}\right)\sum_q K_q.$$

The quark multiplier is a prescribed screening factor. This map includes neither complete charged-vector and scalar response nor a nonperturbative hadronic spectral measure, so it is not the full physical electromagnetic running of the Standard Model.

The minimal census with $\beta = 4$, positive couplings, $P \geq 1$, $M_U \geq m_Z$ and no intermediate thresholds has no solution: the scale equations force $\alpha_2, \alpha_3 > 1/6$, whose combined heat sum is below $1/80$, while pixel closure requires at least $1/4$. With the explicit alternative $\beta = 7$, interval bounds certify a locally unique self-consistent root

$$P \simeq 1.628533369611833, \qquad A \simeq 168.815082928453.$$

Infinite heat tails, implicit-root derivatives and a strict interval self-map are controlled. The memory inverse is certified over the same root enclosure, so its two admissible controllers realize those same charged-lepton coefficients. This is a consistent conditional coefficient map; its inverse coupling is not an identification with the measured fine-structure constant [12].

A complete physical loop must replace the coefficient readout by the current spectral response of the actual interacting states. Compact binding energies and quartic interactions change that spectrum. Therefore closing the displayed scalar map does not close the joined physical particle-response loop. Detector use of $g(P) = \sqrt{4\pi\alpha(P)}$ reads a supplied coupling; it does not independently derive its electromagnetic meaning.

The other closure coordinate, record capacity N, bounds the declared correctable record algebra. Under its de Sitter entropy interpretation,

$$N = \frac{3\pi}{\Lambda \ell_P^2}, \qquad \ell_P^2 = \frac{\hbar G}{c^3}. \tag{15}$$

This dictionary converts a capacity into a cosmological constant after physical scale calibration. It does not make a fitted capacity an independent cosmological prediction. The specialist analyses separate conditional root certificates, measured-coordinate comparisons and independent physical tests [10, 16].

12 Realized results and physical obligations

A statement about nature needs one physical realization carrying compatible events, fields, action, states and readouts across refinement. This section gives, for each layer, what is established on its stated domain, what is a conditional implication, which join the executable model must realize, and which result would stop the layer or its branch. Exact equality of limiting observable laws is the mathematical target; finite executions and experiments are judged with declared approximation and measurement errors.

12.1 Realized finite domains and physical extension

The finite constructions establish exact state, algebra and operation maps on their declared domains. The common particle operator, full-record memory limit, compact-defect energy bounds and noncommuting interacting response supply concrete instances. A list of such instances does not prove that every physical premise is realized on one source.

The scope of the mission is broader than the established physical results. Table 1 pairs each question with its strongest constructed answer and the condition needed to read that answer as a fact about our universe. The entries for existence and subjective experience are interpretive targets. Public measurement laws, including reports of experience, have an operational meaning; the character of an experience is not thereby derived from those reports. The common specification gives these diverse questions a single testable setting. Its physical correctness depends on satisfying the conditions in the right column together, including the measured particle spectrum and responsive geometry, and on surviving independent prospective tests.

Table 1: Explanatory scope and status of the proposed common specification.

Question	**Constructed result**	**Physical or interpretive condition**
Why anything exists	An inhabited self-recovering history is the closure hypothesis; an exact finite consensus closure has fixed packets	A universe-level fixed point and its identification with our history are not established by the finite fixed-point theorem
Why there are public facts	Protected overlap records have canonical normal forms under the displayed repair hypotheses	One source must realize the observer, scheduler and comparison contracts across the claimed domain
Why experience occurs	The executable observer has defined access, memory, intervention and observer-relative record interfaces	These interfaces give no derivation of qualia or of why any record has phenomenal character
Why space has three directions	The twelve-port response has an exact three-dimensional slow band and Euclidean completion	Physical position, length and a cofinal spatial refinement require the realization map (M1)
Why observations are quantum	Completed records obey Born, conditioning and correlation identities in a finite complex algebra-state model	The source must produce that algebra, its tomographic effects and its physical preparations; classical consensus alone does not imply them
Why these forces and particles	Complete-response premises force $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$; a chosen matter module has an anomaly-free generation; native currents, modes and mass coefficients share a finite operator construction	Physical chirality, dynamical gauge fields, stable defect identities and the complete measured mass and mixing spectrum require a common interacting limit
Why fields propagate in $3+1$ dimensions	Controlled free fermion and conditional scalar limits, together with a finite interacting core, have explicit domains	Their states, clocks and interactions need one controlled relativistic continuum limit
Why geometry responds to matter	Null tomography and tensor completion yield Einstein dynamics from the stated common null-balance and entropy premises	The same source must realize the geometry, conserved stress, clock and generalized-entropy response
Why cosmological observables take their values	Finite capacity, response and data comparisons supply conditional relations and tests	A selected physical vacuum, calibrated scales and independent prospective comparisons are required

The physical extension requires autonomous localized excitations, a controlled chiral spatial limit, dynamical gauge fields and a common geometric stress and entropy response. The table summarizes those requirements without treating a finite certificate as empirical identification.

12.2 Results and obligations by layer

Table 2 gives the result and obligation of each layer. "Established" means proved on its stated domain or executed with retained receipts; "conditional" means that the conclusion depends on displayed antecedents beyond its realized finite instances; the open join is what the executable model must supply on the same process; and the stopping test is the result that would remove the

layer, its branch or the tested law. Negative results are findings about the tested law, stated with their exact scope.

Table 2: Established constructions, physical obligations and negative controls.

Layer	**Established and conditional**	**Physical obligation**	**Negative results and stopping test**
0. Closure and agreement	Established: on the finite consensus branch the closure map is an affine idempotent fixing exactly the packets on consensus normal forms; the record-closure equation is posed exactly. Interpretation and acceptance criterion, not a theorem	A history domain and closure operator for the executable model; interpretation maps fixed independently of validation outcomes, compatible across observers and refinement	The bounded completion class does not identify the solution N; no theorem selects a unique universe. A contradiction among indispensable premises stops the proposal
1. Repair mathematics	Established: canonical observable normalizer; consensus normal form under the transactional diamond; adaptive-run, capacity and fixed-federation bounds; Born weights without continuity axiom, Lüders selection, Tsirelson bound, no-copying; positive residual mismatch from fixed holonomy. Conditional: the finite algebra-state representation of completed records is an input	A source-produced complex-tomographically-complete effect and instrument web with full coexistent-effect additivity; an implemented scheduler within the specified process	Scheduler choices remain explicit inputs; deciding consistent extension is NP-complete for succinct Boolean systems; retained routing logs do not identify federation gluing. As theorems they are refuted only by a counterexample
2. Carrier and repair law	Established: the seam rule from four requirements; the classified convex record-law family; the gap $(5-\sqrt{5})/60$ for every admissible table; the rank-three slow-response completion; one seam per native repair event. Conditional: physical position consumes (M1)	One specified record law and federation gluing validated against the physical process; a physical length	At full-port energy scaling the refining screen loses the rank-three band. The C/E instrument test separates C, E and the current-only law; a resolved dispersion outside the band excludes the whole carrier class

Table 2 continued

Layer	Established and conditional	Physical obligation	Negative results and stopping test
3. Executable observer	Established: native observer with record-caused feedback; electromagnetic and scalar detectors; one interacting process with certified infinite-occupation contrast; overlapping observers exact on 5,768 output words; local event clocks; identification of the record law; coherent observers on matched screens; complete $O(1/d)$ error for the smooth four-read protocol; a continuing C/E finite-buffer observer with counted exports and an exact stationary source/complete-record marginal	One continuing autonomous observer with every service, memory, reset and resource accounted; the declared apparatus, schedules and preparations implemented on that same process	Outcome accuracy does not certify the output state; committed native records form a classical process, so correlations beyond $S = 2$ need declared coherent preparation services
4. Fields and clocks	Native screen operator with spectral dimension two; conditional smooth static $(1 + 3)$ scalar readouts; centered eight-valley free fermion limit; ordered gapless spin fluctuation limit	Compatible physical clocks, state and energy normalization; an interacting chiral $3 + 1$ field with the observed massive and massless sectors	Passive repair has no wave sector; screen spectral growth excludes a direct elliptic three-manifold identification; a gapless spin field is not a proved relativistic interacting particle theory
5. Gauge and matter	Conditional Lie-type forcing and finite matter image; common native currents, localized modes and pixel-dependent mass matrix; compact opposite-degree pair; exact interacting finite response	Autonomous finite-energy particles with the observed charges, spin, statistics, mixing and pole masses on the same action and clock	Winding does not pin zero energy; compact binding and quartic interactions shift the spectrum. The scalar pixel root does not close the physical interacting current-response loop
6. Gravity	Established: null tomography and the tensor completion from an explicit null-balance premise; same-Hamiltonian interacting finite Gibbs, energy and noncommuting entropy response. Conditional: the Einstein equation up to Λ under (G1) to (G6)	(G1) to (G6) realized on one refinement tower with the same matter, modular flow and clocks; defect maintenance energy in the same stress	The static radial product fails null balance for its own scalar excitation; stationary repair entropy is distinct from fixed-volume generalized entropy. Failure of (M4) removes the spacetime and Einstein conclusions together
7. Cosmology and tests	Established: the deep-law uniqueness under two premises; $(w_0, w_a) = (-1, 0)$ at fixed capacity; the de Sitter sign; the dispersion band. Conditional capacity and pixel fixed points	A common physical response for P, capacity and scale calibration, and independently fixed observation maps	Minimal matter with the specified scale coefficient four has no pixel root; coefficient seven has a root whose response is not the measured coupling. Exclusion of the band, a C/E rejection, a charged-lepton average beyond three standard uncertainties from the balanced center, or exclusion of $(-1, 0)$ fails its branch

12.3 The realization maps

Five maps name what a physical realization must supply, and each boundary statement in this paper cites them by label.

(M1) Position and scale. An identification of the intrinsic carrier completion with physical position, with a physical length, cofinal overlap gluing between carriers, and a refinement limit (Section 5).

(M2) Clock and energy calibration. A common comparison between internal evolution, laboratory duration and energy in measured units. The source-executed reference clock measures the same controlled intervals as the field, and the corresponding energy observables have complete detectors. Extending and validating that calibration across the physical process remains necessary. Native event counts are a different clock; the rejected direct count-to-optical-phase attachment in Section 13.3 does not reject the calibrated reference clock.

(M3) Laboratory current. An identification of the finite port response with a physical gauge current, with a physical matter action and a continuum operator limit.

(M4) One common physical tower. A single physical refinement family realizing the event, modular, stress, entropy, continuum, vacuum, and scale premises together, rather than several independently declared islands.

(M5) An implemented and matched action. A specified field and interaction law realized by the same native services, with its complete state space, relative normalizations and operational readouts. Parameters may be declared or calibrated; which predictions follow without consuming their measured targets must be stated. Physical identification requires compatible propagation, currents and interactions across refinement.

The finite constructions realize restricted instances of these maps: a calibrated common clock, internal current operators, a full field action, energy readouts and compatible reference observations. A common physical extension and validation, with the requirements of Table 2, is not established by those restricted maps. A theorem that consumes none of them stands on its own finite domain. Section 13.3 tests two simple attachments of native records to published data. The construction paper gives the detailed premise map, including the retained negative controls that bound how the carrier, the response triplets, the feedback class and the kinetic term may be attached [63].

13 Empirical meaning and falsifiability

The custody-bound register fixes target definitions, exclusion thresholds, precision floors, exposure classes and decision rules before eligible comparison [3]. Two kinds of test follow from the construction: a propagation anisotropy fixed by the carrier's symmetry, and an instrument comparison fixed by the executable observer. Published records also test precise attachments of native records retrospectively.

13.1 Propagation anisotropy at angular rank six

Angular rank means spherical-harmonic rank j, not a particle spin. The dimension of the invariant subspace of the sixty proper icosahedral rotations in angular rank j is

$$m_j = \frac{1}{60}\Big[(2j+1) + 15\,U_{2j}(0) + 20\,U_{2j}(\tfrac{1}{2}) + 12\,U_{2j}(\tfrac{\varphi}{2}) + 12\,U_{2j}(\tfrac{\varphi-1}{2})\Big],$$

with U the Chebyshev polynomials of the second kind, and the exact table gives $m_1 = \cdots = m_5 = 0$ and $m_6 = 1$ [51].

Theorem 13.1 (Icosahedral angular-rank universality)**.** *Let the carrier kinetics be any operator $\lambda_a(k) = a^{-2}\sum_d w(d)\,[1 - \cos(a\,k\cdot d)]$ with a finite direction multiset and weights invariant under the proper icosahedral rotation group. Then the dispersion is exactly isotropic through angular rank five at every order in a, every directional term below angular rank ten is one multiple of the unique normalized invariant $\mathcal{I}_6$, and whenever the weighted sixth-moment coefficient is nonzero the first directional artifact appears at order a^4k^6, suppressed by exactly $1/16$ under one binary refinement at that order. Rank six is the least symmetry-allowed nonzero anisotropic rank in the class.*

Proof. The order-$a^{2m-2}k^{2m}$ angular content is an invariant polynomial of degree $2m$ with harmonic components at even ranks $j \le 2m$, and the invariant table forces zero at $L = 2, 4, 8$ and a one-dimensional space at $L = 6$. The table, the 62-direction stationary census, the orbit verification and the nonvanishing sixth-moment coefficients are certified in exact arithmetic [1]. □

Proposition 13.2 (Primitive twelve-port branch prediction)**.** *Suppose the intrinsic spatial kinetic symbol lies in the cosine class of Theorem 13.1 with scale-independent coefficients, the complete orbit of the twelve port directions u_i is its sole hop support through order k^6, the carrier scale a is finite and positive with the quadratic term normalized to k^2, one carrier rest frame is transported coherently over the experiment, and the tested scalar or polarization-independent sector realizes the operator. Then*

$$\omega^2 = \frac{1}{2a^2}\sum_{i=1}^{12}[1 - \cos(ak\,u_i\cdot\widehat{k})] = k^2 - \frac{a^2}{20}k^4 + \frac{a^4}{840}k^6 + \frac{2a^4}{7875}k^6\mathcal{I}_6(\widehat{k}) + O(a^6k^8), \tag{16}$$

with $\mathcal{I}_6(n) = \frac{25}{132}\sum_i P_6(u_i\cdot n)$, so in a transported frame the branch predicts the scale-free relations

$$\frac{B_6}{C_4^2} = \frac{32}{315}, \qquad \frac{B_0}{C_4^2} = \frac{10}{21}, \qquad \frac{B_6}{B_0} = \frac{16}{75}, \tag{17}$$

and every intrinsic anisotropic coefficient at $1 \le j \le 5$ vanishes.

Proof. The exact carrier moments $\sum_i(u_i\cdot n)^2 = 4$, $\sum_i(u_i\cdot n)^4 = 12/5$ and $\sum_i(u_i\cdot n)^6 = 12/7 + (64/175)\,\mathcal{I}_6(n)$ substituted into the cosine series give the expansion; eliminating a gives the ratios. □

A second frozen ray follows from the source seams.

Proposition 13.3 (Source-seam edge ray and conditional propagation branch)**.** *The boundary current of the complete thirty-seam source packet maps through the antipodal-odd load readout onto the even-sum module D_6 of Theorem 5.1, with Smith invariants $(1, 1, 1, 1, 1, 2)$, and every directed seam defines one translation of the internal record carrier whose complete normalized direction multiset is the thirty-direction edge orbit. If the feasible move laws and their objective are natural under the proper carrier action and A3 supplies a unique normalized minimizer, transitivity forces weight $1/60$ on every directed seam, and the selected internal operator is the source-counting homogeneous convolution.*

Suppose these record translations are physical displacements under one homogeneous position action, that the complete edge orbit is the sole direct support through the displayed order, that proper-carrier covariance fixes one common weight, that the same operator acts on one scalar or polarization-independent physical sector with quadratic term normalized to k^2, and that the action

carries the stated gluing, scale, frame, readout, nuisance, and exclusivity data. Its spatial kinetic symbol is then

$$\Lambda_a(k,\widehat{k}) = \frac{1}{5a^2}\sum_{j=1}^{30}\Big[1-\cos\Big(ak\,w_j\cdot\widehat{k}\Big)\Big], \tag{18}$$

with $C_4 = -a^2/20$, $B_0 = a^4/840$ *and* $B_6 = -a^4/12600$, *hence* $B_0/C_4^2 = 10/21$, $B_6/C_4^2 = -2/63$ *and* $B_6/B_0 = -1/15$. *The intrinsic anisotropic coefficients at angular ranks one through five vanish, and the rank-six vector lies on the rotated* $\mathcal{I}_6$ *orbit.*

Proof. Exact integer incidence gives the displayed image and Smith invariants, and the response-selected Gram pullback gives the completion of Theorem 5.1. The sixty directed labels map two-to-one onto the thirty signed edge directions, whose unit-normalized moments are $\sum_j (w_j\cdot n)^2 = 10$, $\sum_j (w_j\cdot n)^4 = 6$ and $\sum_j (w_j\cdot n)^6 = 30/7 - (2/7)\,\mathcal{I}_6(n)$; substitution in (18) gives the coefficients and ratios. The incidence, quotient, orbit binding and moment identities are machine-checked in exact arithmetic [1]. □

The proposition contains an exact finite source theorem and a conditional physical branch. The naturality and minimizer clause supplies a homogeneous internal action whose generator satisfies the positive maximum principle and an exact Dirichlet identity, and whose plane-wave spectrum is the symbol above. The declared symbol also carries an exact remainder certificate with no physical premise: with $q = ak$, $|\widehat{\Lambda} - P_6| \le \frac{7}{388800}q^8$ and $\frac{19}{20}q^2 \le \widehat{\Lambda} \le q^2$ uniformly on $0 \le q \le 1$, where $P_6 = q^2 - q^4/20 + (1/840 - \mathcal{I}_6/12600)q^6$; the certificate turns a into no SI length. The edge branch and the primitive-port branch have opposite rank-six signs, and their coefficient rays, custody packages, exposure exclusions and comparison budgets are separate.

Theorem 13.4 (Carrier-class dispersion band)**.** *For the declared positive-weight scalar cosine class over orbits* O_s *of the sixty proper carrier rotations with radii* $r_s > 0$, *weights* $w_s > 0$ *and the* k^2 *coefficient normalized to one, write* $\Lambda_{\mathrm{norm}} = k^2 + C_4k^4 + (B_0 + B_6\mathcal{I}_6(\widehat{k}))k^6 + O(k^8)$ *and* $\mu_m = \sum_s w_s|O_s|r_s^m$. *Then* $C_4 = -(a^2/20)\mu_4/\mu_2 < 0$ *for every member;* $B_0/C_4^2 = (10/21)\mu_2\mu_6/\mu_4^2 \ge 10/21$ *with equality exactly on single-radius members; the anisotropic ranks one through five vanish; and* B_6/B_0 *lies in* $[-16/135, 16/75]$, *the face and vertex orbits attaining the endpoints and the edge orbit sitting at* $-1/15$.

Proof. Summing over the sixty rotations gives $\sum_g ((gu)\cdot\widehat{k})^6 = \frac{60}{7} + \frac{64}{35}\mathcal{I}_6(\widehat{u})\mathcal{I}_6(\widehat{k})$ on unit vectors. The floor is the Lagrange identity $\mu_2\mu_6 - \mu_4^2 = \sum_{i<j} W_iW_jr_i^2r_j^2(r_i^2 - r_j^2)^2 \ge 0$ with $W_s = w_s|O_s|$, and the band holds because the rank-six numerator is a positive-weighted mean of seed values of $\mathcal{I}_6$ in $[-5/9, 1]$ [1]. □

A resolved intrinsic dispersion with B_0/C_4^2 below $10/21$, or B_6/B_0 outside $[-16/135, 16/75]$, or a residue at ranks one through five, excludes every member of the declared class at once. The primitive-port relation is a frozen prospective conditional physical-branch prediction: its target statement, coefficient snapshot, registration record, public commit custody and detached calendar attestations fix the prospective content, the dated template search whose icosahedral templates returned a family-wide p-value of 0.64 is ineligible, and a verdict requires a dataset-specific registration with a joint likelihood, frames, orientation profile, nuisance model and calibrated threshold. The branch fails at five standard deviations or more if an isolated intrinsic C_4 is positive, a lower-rank coefficient is nonzero, or the calibrated profile likelihood excludes the complete manifold $C_4 < 0$, $B_0 = \frac{10}{21}C_4^2$, $B_6 = \frac{32}{315}C_4^2$. A null is inconclusive, since the branch supplies no positive lower bound on a, and a failure rejects the primitive twelve-port propagation branch; it reaches the framework as a whole only after a stronger source law makes that branch forced and exclusive [3].

13.2 An instrument test from the executable observer

Under a common current-record interface every admissible record law has the classical statistics of C and E. The multiplier g, the coherence between an empty seam and a waiting single load, separates C, E and the law that records only the current. On a fixed coherent two-state source preparation the readout probability is $3/8 + \frac{1}{2}\,\mathrm{Re}(g\,e^{-i\theta})$: reversing the phase moves C from 5/8 to 1/8, E predicts 3/8 for both, and the law that records only the current predicts $3/8 \pm \sqrt{2}/4$ [65]. Four fixed settings include common-phase and incoherent controls. The complete-count evaluator uses 8192 trials per setting, an independently established probability-error budget of 1/64 and an exact sampling certificate below 1/1000; it rejects malformed or incomplete records and reports *no data* when no independent counts are supplied. A source-executed implementation runs both complete 78-state instruments, the four preparations and a complete Lüders readout from twelve-port control words, retaining every record and attempted trial [63]. A sequential design on one continuing carrier needs neither exact resets nor independent trials: 379 discarded native attempts before each of 3,927 trials replace the reset, failed heralds count as trials, and conditional-law bounds on the dependent transcript give a wrong-candidate probability of at most 3/400 under the exact-source model [84]. Synthetic controls exercise the decision rules and establish no physical verdict; physical encoding, independent calibration and comparison data are the open requirements.

13.3 Native records against public data

Published records can test precise attachments of native records without a new laboratory realization; each test below is retrospective, states its readout hypotheses, and assigns no prospective verdict.

The diagonal native branch is a classical stochastic process, so under independently chosen settings and local factorized responses it obeys the CHSH bound $S \le 2$. The deposited records of the loophole-free Hensen experiment [87] reproduce 245 eligible trials, 196 wins and $S = 2.42250$, and the published fixed-count analysis gives a tail bound near 0.03917 with its setting-calibration allowance [85]. This rejects the local classical reading of the diagonal branch under that experimental contract. Committed native records form a classical process: the native grammar keeps diagonal inputs diagonal under every classical feedback history, and two implementations with the same committed branches, which give 3/4 and 1/4 under a read before commitment, give 1/2 for both once the record is committed [93]. Correlations beyond $S = 2$ therefore need uncommitted coherence from declared preparation services, and every admissible record law attenuates it [65]. The replay leaves the completed record surface untouched, on which a declared entangled state reaches the Tsirelson bound of Theorem 4.5, and it does not distinguish C from E.

Event clocks need no external validation, but identifying their readings with a published measurement is a testable hypothesis. Let two native ports, or two distinct regions of one carrier, count incident repairs, and attach the published 300-second blocks of the Ar^{13+}/Yb^{+} optical frequency ratios of King et al. [88] to the normalized count ratio through a gain of at least one, a declared bracket on the Yb^{+} frequency and an absolute error of 10^{-12} per block, with direct phase-count readback. The complete count law, after charging every choice of 2021 subwindow, gain and retained blocks, bounds the probability of the observed block ranges below 10^{-104} for two native ports and below 10^{-22} for every pair of the 3,433 distinct regional event sets [86]. The direct attachment is rejected together with its readout assumptions; the experiment's interrogations and servos lie outside that hypothesis, and the native event clocks are unaffected as clocks of the model.

Published trapped-ion records test a supplied experimental model executed through the source interface. Two calibrated preparation parameters predict 100 reserved complex readouts with per-

spin-mean root-mean-square error 0.06647, and a changed-potential transfer predicts 125 derived observations with root-mean-square error 0.2876 in position-estimator units [64]. These comparisons test the supplied oscillator model and derive neither its action nor its apparatus from native repair.

13.4 Diagnostics and what fails together

The other quantitative comparisons carry their own decision shapes and do not count as predictions. The closure roots of Section 11.2 are diagnostics of declared maps. The registered conditional primordial branch $n_s = 1 - P/48 = 0.96602$, at the calibrated grain P, lies at the lower 0.612 percent position of the official ACT DR6, Planck and DESI joint posterior chains, whose mean is 0.9743 ± 0.0034 [89, 86]. Those data precede the registration, so this is a retrospective tension under baseline ΛCDM, not a likelihood ratio, and the registered comparison with a later dataset is unchanged. A separate ACT DR6 likelihood slice on the measured bandpowers, with the background and absolute calibration held fixed, prefers a free tilt, fitted at 0.9773, by $\Delta\chi^2 = 4.07$; no significance or framework-wide verdict is assigned to it [86]. A charged-lepton world average more than three standard uncertainties from the balanced-circulant center refutes the equal-rank-two tracial reading of the face-circulant sector; a resolved deep-regime galaxy sample violating degree-one radial scale covariance or quadrature composition removes the uniqueness of the deep law; and exclusion of $(w_0, w_a) = (-1, 0)$ fails the fixed-capacity branch. The conditional implications share premises, so they fail in groups: failure of (M4) removes the effective-spacetime and Einstein conclusions together, failure of (M3) removes the gauge and matter interpretations together, and failure of (M2) removes every statement in laboratory units. The finite normal-form, invariant-theory and algebraic results retain their stated domains under all three.

14 Machine verification and reproducibility

The results have different logical types, and Table 3 gives the classes used throughout; each result throughout the paper carries one of them. Each step is identified by its artifact as a Lean theorem, exact executable certificate, conditional paper argument, numerical receipt, or explicit unproved premise.

Each result maps to an artifact class: a Lean library with no admitted propositions in the dependency closure reported here, including formal premise boundaries and countermodels; exact-arithmetic code receipts, integer and rational computations and outward-rounded interval certificates, each emitting a canonical hashed payload; executable observer receipts that replay every retained result and check its artifact hashes; simulation receipt bundles pinning the source revision, configuration, seed where stochastic, tolerances and output hashes; verifiers that recompute verdicts from clause vectors and reject receipts whose stored verdict disagrees; and adversarial negative controls, so that every certificate ships with mutations that must be detected. Machine-readable records classify every quantitative row as a source result, reconstruction implication, diagnostic, prospectively fixed test or rejected candidate, and reject any stronger classification unsupported by the recorded dependencies.

Table 4 names the artifact family behind each group of results. The repository, receipts, schemas and rebuild instructions are public [1, 2, 64]. The formal and executable stack checks the claims assigned to it and enforces their stated boundaries; analytic arguments in this manuscript retain their displayed proofs, and no part of the stack establishes physical realization.

Table 3: Epistemic claim classes used in the manuscript.

Finite theorem	A deductive result proved from displayed premises on a stated finite domain. An exhaustive computation counts as part of the proof only with a specified domain and a soundness argument.
Certified finite computation	An executable witness whose inputs, controls, and output are fixed by a reproducible certificate; execution alone does not promote its interpretation to a theorem.
Conditional analytic theorem	A theorem whose hypotheses include a supplied continuum action, state space or limit; its conclusion holds on that supplied object.
Source limit theorem	A theorem about the limit of a declared family of finite source executions along a stated subsequence, with its normalizations and order of limits displayed; the limit is derived, and its physical identification is a separate claim.
Reconstruction implication	An exact implication whose antecedents include displayed geometric or physical premises.
Unsupplied realization map	A required map from a finite object to a physical event, current, field, clock, scale, or continuum structure for which no construction is asserted.
Diagnostic	A certified computation compared against measured values that it consumes; never counted as a prediction.
Prospectively fixed conditional test	A diagnostic with a custody-bound target definition, kill band, precision floor, and decision rule fixed before comparison.
Frozen prospective branch prediction	A source calculation and decision rule fixed before any eligible comparison, conditional on a named physical branch. Failure rejects that branch; its scope extends to the whole framework only if the branch is proved forced and exclusive.

Table 4: Principal verification artifacts and their scope.

Results	**Principal artifacts**
Executable observer, detectors, interacting process and C/E instruments	The observer receipts with exhaustive path replays, seeded executions, feedback controls, exact and interval certificates, and a single verification entry point that replays every receipt and checks its hashes [64, 63]
Overlapping observers, local clocks, screens and the continuum operator	The model packages for the observer correspondence, the compiled clock history, joint and active clocks, repair identification, federated screens and the coherent screen observer, each with exact or interval receipts, independent oracles and replay; the analytic limit arguments with their constants and order of limits [73, 80]
Public-data comparisons	Hash-pinned public records (Bell trials, optical ratio traces, CMB posterior chains) with exact rational reducers, their stated readout hypotheses and replay checks [85, 86]
Normal forms, stability, refinement, complexity and the transactional diamond	The observable-normal-forms library and component-paper reductions; the local-diamond and Newman arguments of this manuscript; the finite transaction-model verifier
Born weights, Lüders selection, Tsirelson saturation, no-cloning	The event-algebra modules for the interface, conditioning, the bound and the public record algebra, with per-module axiom audits
Rank-three completion	The port-Gram repair band, covariance and frame-quotient modules with a source-pinned receipt and independent verifier
Gauge Lie type, matrix witness, matter image and exterior selection	The screen modules for the carrier action, commutant and incidence response; the exact-arithmetic port current certificate; the trace-balanced kernel and exterior scan modules
Causal precursor, count limit and proper-time readout	The provenance and geometry modules; the source-record family receipts with exact all-pairs counts through $q = 89$ and the sampled $q = 144$ level, the carrier realization receipts and event logs, and the public archive
Null tomography, small-ball arithmetic and tensor completion	The Einstein-branch tensor module for null directions, design, decoder, injectivity and metric ambiguity; explicit null-balance, Ward, Bianchi and scale premises
Angular-rank theorem, frozen branches and the dispersion band	The primitive-port prediction and carrier-class band modules with mutation tests; the custody-bound prediction register
Exact federation and the closure loop	The level-six configuration, seeds, seam arrays, loads, per-schedule ledgers, quotient hashes, standalone verifier and public archive; the level-ten settlement receipts; the event logs and recovered specifications of the canonical and control sources
Dark sector, vacuum coordinate and closure roots	The dark-sector, deep-profile, per-cut and fixed-capacity modules with the matched seen-data diagnostic; the capacity closure and nonidentifiability modules; the interval-certified grain root

15 Discussion and comparison

Operational reconstructions derive quantum structure from information and composition principles [19, 20, 21]. OPH makes a narrower statement at the quantum stage: it takes the finite algebra-state representation as an input and contributes the completed-record surface on which the event, conditioning and correlation identities operate, together with the consensus theorem that defines completion. Quantum Darwinism locates classical objectivity in redundant environmental records [46], and the completed-consensus record plays the same public role with redundancy replaced by

protected overlap agreement and an explicit repair dynamics. The cellular-automaton programme shares the finite substrate [47], while OPH locates the quantum identities on the completed record surface. Wigner's-friend extensions constrain agents reasoning about one another's unfinished measurements [48], whose premises concern pre-consensus perspectives that the completed surface does not represent. Relational quantum mechanics treats physical facts as relational [44] and the thermal-time programme reads time from modular structure [45]; OPH attaches public facts to protected overlap records, and its executable observer reads modular flow from the actual observation state of a running process and lets native incident events drive a modular step whose generator is the native transfer, with a faithful thermal reference derived from closed native histories; passive repair alone implements no nontrivial modular automorphism. The stationary native state of a fixed-number sector is tracial, so its modular flow is trivial, while conditioning on a native current event gives a Palm reference whose coherent modes carry modular frequency $\log(5/4)$, the ratio of the ten moving branches of a non-adjacent occupied pair to the eight of an adjacent one, a degree-census identity of the carrier: modular time in this construction is relative to an event [64]. The local-diamond proof sits beside asynchronous agreement [40, 41], and Dijkstra's self-stabilization separates recovery from continued legitimate operation [42].

The gravitational branch belongs to the thermodynamic and entanglement-based routes to geometry [55, 56, 59], with Bisognano–Wichmann modular flow and half-sided modular inclusions as established analytic ingredients [52, 53, 54]. The route from a null-projected balance and the Bianchi identity to the full equation, with Λ as an integration constant, is familiar from trace-free and unimodular formulations [57, 58]; what is added is the finite nine-direction design with exact constants and the premise contract (G1)–(G6) stated as one implication. The gauge section uses Doplicher–Roberts reconstruction for the sector group [61, 62] and contributes the forcing of the local Lie type from complete response and rechartings inner on that response, a stronger premise than mere internal availability of their implementers.

The closure hypothesis has precedents of a different kind. Wheeler pictured a participatory universe as a self-excited circuit [95]. Tegmark's mathematical-universe hypothesis identifies physical existence with mathematical structure [96], and Nozick proposed that a fundamental principle might subsume itself [97]. OPH adds two demands: the self-subsumption must be inhabited, carried out through records and agents inside the structure, and it must be testable in part, through an executable specification whose observers can be run and whose instruments make predictions. Lawvere's fixed-point theorem shows why unrestricted self-representation fails [98]; the closure condition is restricted to the observable and interventional structure that observers can recover.

A useful comparison of unification proposals asks how much physical structure follows from each declared input and which observations distinguish the resulting models; OPH offers a concrete case with proofs and executable checks attached to individual implications. Formal verification measures deductive reliability, predictive success requires physical realization and eligible measurements, and neither theorem counts nor numerical coincidences define a probability that an architecture describes nature.

16 Conclusion

OPH asks for one executable specification whose compatible observer records reproduce fields, particles and gravity. A simulator design may be recovered and implemented; its consequences must then be derived without silently changing the state, operators, clock or parameter ancestry between sectors. Physical completeness means reproducing the observed joint laws and intervention responses, not deciding every sentence of a formal language.

The finite theorems establish public record normal forms, quantum record identities, a rank-three slow carrier response, and conditional forcing of $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$. The specified matter module has one anomaly-free fifteen-state generation and a faithful $\mathbb{Z}_6$ quotient. Native current operators, a pixel-dependent lepton matrix and localized fermion modes have an explicit common finite realization. Unread final states have an $O(1/n)$ memory limit, and the complete final history has an $O(n^{-1/2})$ comparison retaining its actual conditional law. A nonquadratic extension has exact connected correlations and shifted spectral poles on stated reducing sectors.

The spatial limits have precise domains. The native screen gives a local Gaussian operator of spectral dimension two; a chosen radial service admits smooth static $(1+3)$ scalar descriptions on finite observation menus. The centered Weyl/contact refinement gives eight free massless valleys for $0 \leq U = o(c^{2/7})$ as $c = v/a \to \infty$. A distinct ordered spin trajectory gives a gapless, nonstatic fluctuation field, without a proof of relativistic dispersion or surviving continuum interaction. The finite null-balance theorem yields Einstein dynamics when the same geometry, stress, clocks and generalized-entropy stationarity instantiate its premises. These results do not compose into a controlled interacting chiral theory with responsive Einstein geometry by assertion.

Two consistency requirements are explicit. A compact defect pair has finite energy bounds, but its localized energies shift the radial mass coefficients; interactions shift them again. The physical pixel loop must use the current response of those same states. Its conditional scalar root certificate alone supplies neither measured particle identity nor a physical Thomson limit. Autonomous finite-energy defects, their observed quantum numbers, the complete matched spectrum and the common gravitational response are therefore unproved physical conclusions.

At the broadest level, closure offers a candidate answer to why there is an inhabited world. The observer construction defines access to records and their public agreement; it gives no theorem about phenomenal character. A complete account must join that interpretive question to the physical requirements of Table 1.

The frozen carrier dispersion band and complete-record instrument test provide falsifiable statements on their stated domains. Their verdicts must retain their model class and calibration assumptions. Finite consistency, empirical adequacy and physical completeness are separate requirements on the same proposed world model.

Reproducibility statement

All theorems, certificates, simulation configurations, receipts, and the formal library are public in the project repositories [1, 2]. The paper release manifest records the release identifier (r2060) with PDF hashes and sizes; computational receipt bundles pin the applicable source revision, configuration, seed and output hashes, with clean-checkout rebuild instructions and negative controls; and machine-readable claim and prediction provenance accompanies the artifacts [3]. This manuscript synthesizes and strengthens results whose longer proofs also appear in public component preprints on observable normal forms, finite event algebras, consensus, Einstein reconstruction, gauge structure and the executable observer [4, 5, 6, 9, 11, 63], a relationship disclosed here so that overlap is visible.

Statements and Declarations

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Author contributions. Bernhard Mueller developed the OPH concept and mathematical synthesis, wrote the main manuscript, and coordinated the formal and computational evidence. Mario Poneder developed the executable observer construction, including the native observer, the interacting source process and their certificates, and contributed the finite port-current construction and verifier, gauge and matter certificates, and proof review of the edge-entropy and Einstein-normalization branches. Alexander Osika proposed that the screen pixel ratio and the correctable-record capacity are derivable rather than fitted, supplied the detuning fixed-point equation that determines the pixel ratio, and originated the echosahedral patch-hardware design on which the screen-microphysics branch is built, in addition to the physical-realization and hardware boundary, prototype framing, and manuscript review. Kai Xue contributed consistency checks, simulator review, visualization review, and prototype-engineering review. Ben Cassie contributed implementation-claim and physical-evidence review. Peter Nguyen contributed proof auditing, branch-boundary analysis, compact-gauge and global-quotient analysis, and manuscript consistency review. Jinwook Kim developed the constructions, proofs, and counterexamples for the profile partition and structural characterizations, exact behavior-cut recovery, exact finite-state profile transport, adaptive repair stratification, the cumulative attempt-capacity classification, fixed computation-federation progress, and the fixed-federation execution classification, together with the mechanism-variant comparison kernel and conditional example and the source-derived phase-effect selection. He constructed a provably fair emission-order round-robin scheduler with a linear exact-output attempt bound, established order-sharp quadratic accepted-step bounds for the actual compiler with the exponential fanout control, and proved the historical weak-relation fair-stuttering no-go and the weak-fair delay versus bounded-waste horizon separation. He formalized these results in Lean, connected the repair and execution results to the Tower library, and contributed their manuscript, claim registry, ledger and build records. David Matscheko contributed proof review of consensus repair, observable normal forms, modular and Einstein algebra, edge entropy, hypercharge, and the $\mathbb{Z}_6$ quotient. Daniel Ari Friedman contributed independent red-team audits of the papers, the book, the claim and premise ledgers, the Lean library, continuous integration and the custody records, which led to scoped headline claims, displayed values matched to their certified receipts and repaired falsification rules, and the independently reviewed programme-obstruction audit whose controls bound how the carrier, the response triplets, the feedback class and the kinetic term may be attached. Jonathan Hill contributed Lean formalization and proof auditing of observable normal forms, refinement and repair results, complexity classifications, and artifact coverage. William T. Glynn contributed simulator development, computational implementation, reproducible-build infrastructure, and release-validation work in the main OPH research repository. Maarten Antonie Visser contributed consensus-protocol review, physics feedback on the foundations and emergent-spacetime framing, and the source-derived causal-poset reading of authenticated provenance that the event-precedence layer implements. Kale Arnav Anirudha contributed consensus-paper material, asynchronous-agreement analysis, repair-map definitions, and formal review of the protocol surface. Brieuc de La Fournière contributed finite Lean constructions on the source-derived causal-order and carrier-symmetry surfaces, including the cover-relation provenance compiler and the antipodal port-action identification, and external comparison controls for the charged-lepton branch. All authors reviewed and approved the manuscript. All authors agree to be accountable for all aspects of the work.

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Data availability. The finite witnesses, simulation configurations, receipts, scientific claim registries, and release manifests supporting this study are available in the public OPH and simulation repositories [1, 2]. Individual receipts and their bundle documentation provide the source

and output provenance for computational claims. The artifact map identifies the level-six exact-federation archive and its level-five to level-ten extension, the closure-loop archive, and the causal-poset archive.

Code availability. The Lean sources, exact-arithmetic verifiers, finite scans, simulation code, tests, and rebuild instructions are available in the same public repositories [1, 2]. The computational claims in this manuscript identify their principal files in Section 14.

Ethics approval and consent to participate. Not applicable. The study involved no human participants, human data, or animals.

Consent for publication. Not applicable.

Competing interests. The authors participate in OPH-related research, software, simulation, or prototype-development programs and may receive professional or reputational benefit from this work. These relationships are disclosed. The authors declare no other competing interests.

Credited prior work

The results of this paper build on the following established work. Each entry names an established result as it is used here and cites its sources.

- Icosahedral harmonics, finite-difference dispersion, and nonminimal Standard-Model Extension coefficient searches [106, 108, 104, 105].
- Root lattices, graph incidence, transverse projectors, free-vector oscillators, icosahedral edge harmonics, finite-difference symbols, Lorentz-violating threshold kinematics, and nonminimal Standard-Model Extension coefficient searches [100, 102, 109, 107, 103, 104, 105].
- Spherical-design moment theory, lattice dispersion expansions, and Lorentz-violation coefficient parametrizations [101, 107, 104, 99].

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